Leveraged Learning

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This work rests on three insights.

1. Learning is the incremental ability to answer questions correctly that follow a pattern.
2. The number of questions reviewed is not the right measure of a learner's progress. What counts is the information received from asking them. Not all questions are created equal: about some, the learner already knows a great deal a priori, and their answers count for less.
3. One can infer *more* than a single bit from a yes–no question, by exploiting the correlations of an intelligent prior: an answer over here settles expectations over there. This ability to extrapolate is codified here as *leverage*.

1 A thought experiment

Let us start with a small thought experiment, by introducing *Werner the learner* (Figure 1).

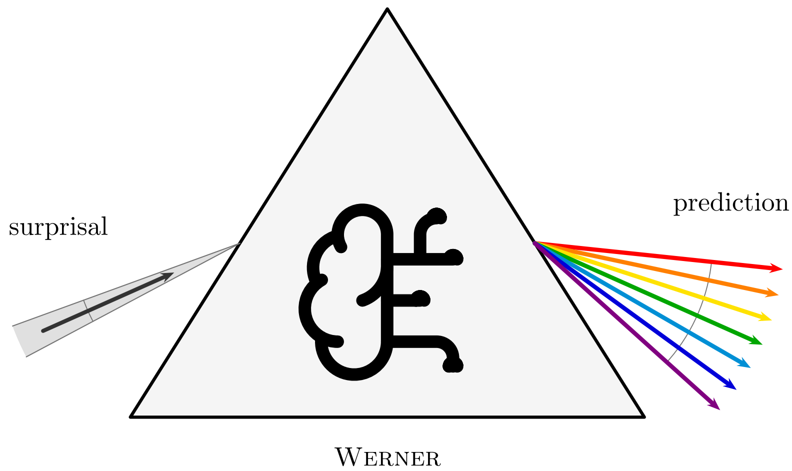


Figure 1: Werner the learner. One thin ray of surprisal enters his prior and leaves spread across a broad fan of prediction: the optical image of leverage, in which a narrow input is refracted into a much wider output.

In this work, we model learning as the increased ability to give the correct answer to questions, as more of the ground truth is revealed. It is Werner's job to learn a pattern. The most basic shape of a pattern is a *binary map*, and the most basic binary map is a map from one bit to one bit.

The cast of the experiment:

- a **question** $q \in \mathcal{Q} = \{0, 1\}$;

- an **answer** $a \in \mathcal{A} = \{0, 1\}$;
- the **truth** $\psi : \mathcal{Q} \to \mathcal{A}$, the one fixed map nature uses to answer questions. ψ is unknown to Werner. By definition then, $\psi(q) = a$.

There are exactly four candidate maps ϕ_j that ψ could be (Table 1): each of the two inputs can be sent to either of the two outputs, independently. We index them by reading the truth table $\phi_j(1)\,\phi_j(0)$ as a binary numeral: that number *is* j.

j	$\phi_j(1)$	$\phi_j(0)$	name
0	0	0	constant 0
1	0	1	NOT ($\neg q$)
2	1	0	identity (q)
3	1	1	constant 1

Table 1: The four binary maps from one bit to one bit.

Now, if Werner is unintelligent, he has no preconceived notion of which maps are more useful, more likely, or somehow more logical than others. In practice, we would have to tell Werner the answer a to every single question q, before he has learned the whole pattern.

We model this with a **uniform prior**: writing $p_j = \Pr(\psi = \phi_j)$ for Werner's belief, every candidate map carries the same weight.

EXAMPLE

For the four maps at hand:

$$p_j = \tfrac{1}{4}, \qquad j = 0, 1, 2, 3. \tag{1}$$

What does this belief predict? For each question q separately, we can tabulate the probability of each possible answer a: it is the total belief in the maps that would answer a to q,

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j. \tag{2}$$

Arranged with one row per answer ($a = 0, 1$ from the top) and one column per question ($q = 1, 0$ from the left, descending like the digits of the numeral convention), these numbers form the matrix M - each column a probability distribution over answers, so M is column-stochastic.

EXAMPLE

Under the uniform prior, two of the four maps answer 0 and two answer 1, at either question:

$$M \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & M_{0,1} & M_{0,0} \\ \mathbf{1} & M_{1,1} & M_{1,0} \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{3}$$

Every column is a fair coin: the unintelligent Werner predicts nothing about any answer before it is revealed.

Next, Werner gets to predict the answer to the question $q = 0$. By his own belief it is a coin toss: $M_{0,0} = M_{1,0} = \frac{1}{2}$. Let us say the answer comes back $a = 1$. The **surprisal** of this answer, to Werner, is 1 bit, because the probability he assigned to it was one half.

In general, when an event a belief holds with probability P actually occurs, its surprisal is

$$s \;=\; -\log_2 P \quad \text{bits}, \tag{4}$$

here $s = -\log_2 M_{a,q}$ for answer a arriving at question q. Surprisal measures how much the received information deviates from what was already expected: an answer held certain ($P = 1$) carries 0 bits — nothing new; a coin toss carries exactly 1 bit; and the surprisal grows without bound as the answer gets more unexpected, $P \to 0$. It is additive: the surprisal of two independent events occurring is the sum of their surprisals, just as their probabilities multiply.

Moreover, Werner can use a process of elimination — an extreme form of Bayes' rule — to update his prior into a posterior. Semantically: every candidate map that would have answered the question wrongly is now *ruled out*, not merely disfavored.

EXAMPLE

The truth ψ answered 1 at $q = 0$, and a map with $\phi_j(0) = 0$ cannot be ψ. This kills the constant-0 map ($j = 0$) and the identity ($j = 2$). The two survivors, NOT ($j = 1$) and constant 1 ($j = 3$), both predicted the observed answer, so this round of evidence does not distinguish between them at all (Table 2).

j	$\phi_j(1)$	$\phi_j(0)$	name
~~0~~	~~0~~	~~**0**~~	~~constant 0~~
1	0	1	NOT ($\neg q$)
~~2~~	~~1~~	~~**0**~~	~~identity (q)~~
3	1	1	constant 1

Table 2: The candidate maps after the answer $\psi(0) = 1$: the two maps with $\phi_j(0) = 0$ are ruled out.

The surviving ratio of believabilities must be preserved, and the only update that does so is to renormalize the surviving weights to sum to one. (This is Bayes' rule with a 0/1 likelihood: $\Pr(\psi = \phi_j \mid a) \propto \Pr(a \mid \phi_j)\, p_j$, and $P(a \mid \phi_j)$ is 1 for a map that answers a and 0 for a map that does not.)

EXAMPLE

The posterior is

$$p' \;=\; \left(0,\ \tfrac{1}{2},\ 0,\ \tfrac{1}{2}\right). \tag{5}$$

Constructing the new answer matrix from p' by the same recipe as before:

$$M' \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p'_0 + p'_1 & p'_0 + p'_2 \\ \mathbf{1} & p'_2 + p'_3 & p'_1 + p'_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & 0 \\ \mathbf{1} & \frac{1}{2} & 1 \end{array}. \tag{6}$$

So indeed, Werner knows the answer to $q = 0$ — that column has collapsed onto $a = 1$ — and he is still as agnostic as possible about the other answer: the $q = 1$ column remains a fair coin. He gained 1 bit of surprisal information, and cleaned up his table to the tune of 1 bit: the $q = 0$ column went from a full bit of answer-uncertainty to none, while the $q = 1$ column is unchanged.

After the next question, $q = 1$, he learns $a = 1$ — again a coin toss beforehand ($M'_{1,1} = \frac{1}{2}$), so again worth 1 bit of surprisal. Elimination now rules out NOT ($\phi_1(1) = 0$), leaving a single survivor:

$$p'' = \big(0,\ 0,\ 0,\ 1\big), \qquad M'' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 0 & 0 \\ \mathbf{1} & 1 & 1 \end{array}. \tag{7}$$

So this is the constant map: $\psi = \phi_3$. Werner is done. Every column is one-hot, and he will be able to answer all questions correctly from now on.

2 Intelligent priors

Now, let us assume that Werner knows *something* about the world before he starts.

EXAMPLE

Let us say Werner knows the map is not constant. His belief then spreads evenly over the two remaining candidates, NOT ($j = 1$) and the identity ($j = 2$):

$$p = \big(0,\ \tfrac{1}{2},\ \tfrac{1}{2},\ 0\big). \tag{8}$$

Is he any better at predicting $\psi(0)$? Constructing M by the usual recipe:

$$M = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{9}$$

He is still just as bad as before: every column a fair coin.

So what has all this veteran wisdom about the nature of patterns in the world bought him? His increased ability for *inference*.

EXAMPLE

Ask $q = 0$ again, and let the answer again come back $a = 1$ — again a coin toss beforehand, so again worth exactly 1 bit of surprisal. Elimination rules out the identity ($\phi_2(0) = 0$), and the lone survivor is NOT:

$$p' = \big(0,\ 1,\ 0,\ 0\big), \qquad M' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1 & 0 \\ \mathbf{1} & 0 & 1 \end{array}. \tag{10}$$

This time *both* columns collapse at once. Werner received the same single bit of surprisal as before, but cleaned up 2 bits of table: the answer at $q = 0$ he observed, and the answer at $q = 1$ he *deduced*, without ever asking — ruling out the constants means $\phi(1) \neq \phi(0)$, so his prior held the two answers perfectly anticorrelated, and settling one settles the other. He is done after one question instead of two.

We say the information from the first answer was **leveraged**. We will make this more precise in the upcoming sections.

2.1 A one-parameter world

For more structure, let us relax the claim that there are *no* constant maps. Instead, we create a one-parameter world where the prior probability that the map is constant is p, spread evenly within each class (Table 3):

j	$\phi_j(1)$	$\phi_j(0)$	name	p_j
0	0	0	constant 0	$p/2$
1	0	1	NOT ($\neg q$)	$(1-p)/2$
2	1	0	identity (q)	$(1-p)/2$
3	1	1	constant 1	$p/2$

Table 3: The one-parameter prior: constant with probability p.

In vector form, and with the answer matrix constructed by the usual recipe:

$$\big(p_0,\ p_1,\ p_2,\ p_3\big) \;=\; \tfrac{1}{2}\big(p,\ 1-p,\ 1-p,\ p\big), \qquad M \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}\,, \tag{11}$$

fair coins in every column, for every value of p.

How much is there to learn, in total? We account for the table as a whole by declaring entropy *extensive* over the columns — the answer-uncertainties of the columns simply add:

$$H(M) \;:=\; \sum_q H\big(M_{\cdot,q}\big), \qquad H\big(M_{\cdot,q}\big) \;:=\; -\sum_a M_{a,q}\,\log_2 M_{a,q}. \tag{12}$$

Here both columns are fair coins carrying 1 bit each, so the table starts at its maximum: $H(M) = 2$ bits.

Now let us ask a question. Again we could use $q = 0$; by the symmetry of the prior it actually does not matter for this argument. And we are now going to talk about *expected* values, instead of realized ones: averages over the possible maps, denoted $\langle\cdot\rangle$. Since we assume our prior is *a priori* correct — nature draws the truth from the very distribution Werner believes — the average over ψ simply zips with the weights: $\langle X\rangle = \sum_j p_j\, X(\psi{=}\phi_j)$.

How much surprisal should Werner expect from asking a question q? The answer comes back a exactly when the truth answers a, which under the zipped average happens with probability $\sum_{j:\phi_j(q)=a} p_j = M_{a,q}$ — the very probability Werner assigned to it — and when it does, it costs him $-\log_2 M_{a,q}$ of surprisal. Grouping the average over maps by the answer each map gives:

$$\langle s_q\rangle \;=\; \sum_j p_j\,\big(-\log_2 M_{\phi_j(q),q}\big) \;=\; \sum_a \underbrace{\sum_{j\,:\,\phi_j(q)=a} p_j}_{M_{a,q}}\,\big(-\log_2 M_{a,q}\big) \;=\; \sum_a M_{a,q}\,\big(-\log_2 M_{a,q}\big). \tag{13}$$

The right-hand side is exactly the column entropy $H(M_{\cdot,q})$ of equation (12): **the expected surprisal of a question is the entropy of its column**. Each answer occurs with exactly the probability Werner assigned to it, so his average surprisal is his own uncertainty about that answer — an identity that will do a great deal of work in what follows.

For a two-answer column, this entropy is a function of a single number, the probability x of one of the two answers: the **binary entropy function**

$$h_2(x) \;:=\; -x\log_2 x \;-\; (1-x)\log_2(1-x), \tag{14}$$

zero at $x \in \{0,1\}$ (a settled answer) and maximal, 1 bit, at the fair coin $x = \frac{1}{2}$. For the first question here the column is a fair coin, so

$$\langle s_1 \rangle \;=\; H\big(M_{\cdot,0}\big) \;=\; h\big(\tfrac{1}{2}\big) \;=\; 1 \text{ bit}, \tag{15}$$

whatever the value of p.

And the other column is now cleaned up. Breaking the update down over the possible maps: with probability $\frac{1}{2}$ the answer is 1 (truth is NOT or constant 1), with probability $\frac{1}{2}$ it is 0 (truth is FALSE or the identity), and

$$M' \;=\; \underbrace{\begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1-p & 0 \\ \mathbf{1} & p & 1 \end{array}}_{a_1=1,\ \text{prob } 1/2} \quad \text{or} \quad \underbrace{\begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p & 1 \\ \mathbf{1} & 1-p & 0 \end{array}}_{a_1=0,\ \text{prob } 1/2}. \tag{16}$$

Either way, the asked column is one-hot and the unasked column is a coin of bias p: what survives on $q = 1$ is precisely the class uncertainty — constant or not — and nothing else. The total entropy of the table is now

$$H(M') \;=\; 0 + h_2(p) \;=\; h_2(p), \tag{17}$$

a reduction of $2 - h_2(p)$ bits, purchased with a single bit of surprisal. Define the **leverage after one question** as the ratio of table entropy destroyed to surprisal received:

$$L_1 \;:=\; \frac{2 - h_2(p)}{1} \;=\; 2 - h_2(p) \;\in\; [1,2]. \tag{18}$$

The expected surprisal on the second question is, once more, the entropy of its column,

$$\langle s_2 \rangle \;=\; h_2(p) \;\leq\; 1 \text{ bit}, \tag{19}$$

after which the table is empty. Cumulatively over the two questions, $1+h_2(p)$ bits of received information reduced the table by the full 2 bits it started with, so the **cumulative leverage** after two questions is

$$L_2 \;:=\; \frac{2}{1 + h_2(p)} \;\in\; [1,2]. \tag{20}$$

So our knowledge about the world has allowed us to be more certain after our inference. Set $p = \frac{1}{2}$, and the prior is the uniform one of Section 1: $h = 1$, $L_1 = L_2 = 1$, and nothing is leveraged at all. As p skews further from $\frac{1}{2}$, the second question carries less and less expected information, and Werner's potential for learning from it goes down with it. The column it could clean up is nearly settled already. In the limit $p = 0$, we recover the non-constant world from the start of Section 2. There is no surprisal and no improvement to M left in the second question at all: it contributes nothing to denominator or enumerator, and $L_1 = L_2$. Figure 2 traces all three curves between these extremes.

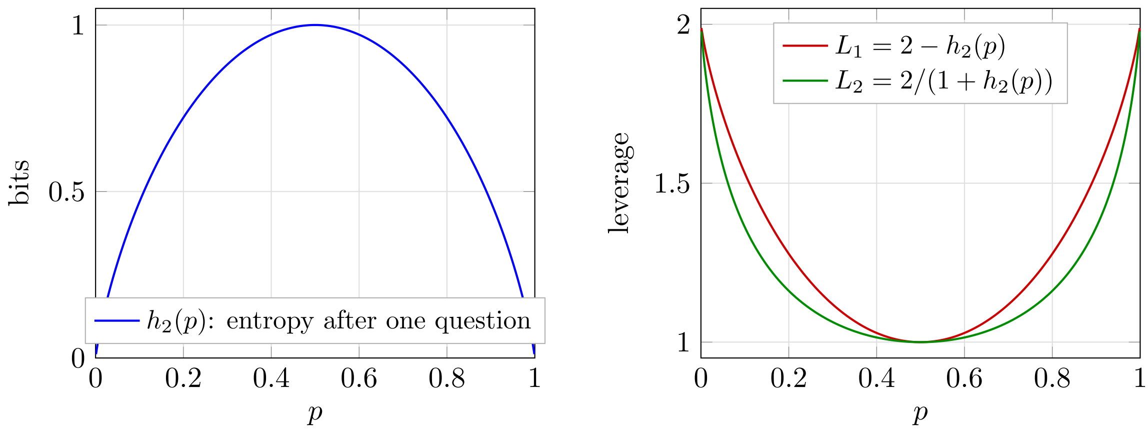


Figure 2: The one-parameter world as a function of the constant-map probability p. Left: the table entropy $h_2(p)$ remaining after one question, in bits. Right: the one-question leverage L_1 and the cumulative leverage L_2 after both questions. At $p = \frac{1}{2}$ (the uniform prior) nothing is leveraged; toward the deterministic-class extremes every answer doubles its value.

In general, in a larger world, if there are d questions to disagree on between the two maps, the reader can convince themselves that the limit as $p \to 1$ is $L_1 = d$.

3 The general case

Now let us give the definitions in full generality. Questions are bit strings of length n, and answers are bit strings of length m:

$$q \in \mathcal{Q} = \{0,1\}^n, \qquad a \in \mathcal{A} = \{0,1\}^m, \tag{21}$$

and we write the sizes of those two sets without bars,

$$Q \;:=\; |\mathcal{Q}| \;=\; 2^n, \qquad A \;:=\; |\mathcal{A}| \;=\; 2^m, \tag{22}$$

so there are Q distinct questions and A possible answers to each. As before, we read bit strings as the numbers they spell, using the two interchangeably: $q \in \{0, \dots, Q{-}1\}$ and $a \in \{0, \dots, A{-}1\}$. The truth $\psi : \mathcal{Q} \to \mathcal{A}$ is one of the candidate maps ϕ_j; since each of the Q inputs can be sent to any of the A outputs independently, there are

$$N \;=\; A^Q \;=\; 2^{mQ} \tag{23}$$

of them, and the index j is again the truth table read as a numeral, now with Q digits in base A: $j = \sum_q \phi_j(q)\, A^q$, running from the constant-0 map ($j = 0$) to the constant-($A{-}1$) map ($j = N{-}1$). Werner's belief is a distribution over all of them, $\sum_{j=0}^{N-1} p_j = 1$.

EXAMPLE

The example we will use to guide the general case is $n = 2$, $m = 1$, which has $N = 2^{1 \cdot 4} = 16$ maps: every Boolean function of the two input bits b_1, b_0, with $q = b_1 b_0$ read as an ordinary binary number, $q = 2b_1 + b_0$. Each ϕ_j in Table 4 is shown as its truth table, printed in descending question order $(11, 10, 01, 00)$ so that the answer to question q occupies the digit of weight 2^q. The printed string is exactly j in binary (e.g. AND, which answers 1 only to $q = 11$, reads 1000, so it is $j = 8$). The prior p_j alongside is conjured for illustration.

j	$\phi_j(q)$ 11	10	01	00	name	p_j
0	0	0	0	0	FALSE (constant 0)	0.394
1	0	0	0	1	$\neg(b_1 \vee b_0)$ (NOR)	0.010
2	0	0	1	0	$\neg b_1 \wedge b_0$	0.010
3	0	0	1	1	$\neg b_1$	0.010
4	0	1	0	0	$b_1 \wedge \neg b_0$	0.131
5	0	1	0	1	$\neg b_0$	0.010
6	0	1	1	0	$b_1 \oplus b_0$ (XOR)	0.010
7	0	1	1	1	$\neg(b_1 \wedge b_0)$ (NAND)	0.010
8	1	0	0	0	$b_1 \wedge b_0$ (AND)	0.213
9	1	0	0	1	$b_1 = b_0$ (XNOR)	0.092
10	1	0	1	0	b_0 (projection)	0.040
11	1	0	1	1	$b_1 \to b_0$ (IF THEN)	0.010
12	1	1	0	0	b_1 (projection)	0.010
13	1	1	0	1	$b_0 \to b_1$ (IF THEN)	0.030
14	1	1	1	0	$b_1 \vee b_0$ (OR)	0.010
15	1	1	1	1	TRUE (constant 1)	0.010

Table 4: The guiding example: all 16 maps of the $n = 2$, $m = 1$ hypothesis space, with a contrived prior ($\sum_j p_j = 1.000$).

The answer matrix is defined by the same rule as before — the total belief in the maps that would answer a to q:

$$M_{a,q} = \sum_{j\,:\,\phi_j(q)=a} p_j, \tag{24}$$

now a $A \times Q$ matrix, still with each column a probability distribution over the possible answers: column-stochastic.

It is worth taking a moment to describe the constitution of this matrix. Because of the enumeration of j (the index *is* the truth table) the answer matrix has a natural structure: it is the contraction of the belief vector p with a fixed tensor with $\{0,1\}$ entries E, summed along the map index,

$$M_{a,q} = \sum_{j=0}^{N-1} E_{a,q,j}\, p_j, \qquad E_{a,q,j} := \delta\big(\phi_j(q),\, a\big) = \delta\Big(\big\lfloor j \cdot 2^{-qm} \big\rfloor \bmod A,\ a\Big). \tag{25}$$

The second form is pure arithmetic on the index: by the numeral convention, $\phi_j(q)$ is nothing but the q-th digit of j in base A, so E never needs to be stored. Membership of j in the entry $M_{a,q}$ is read directly off the digits of j. This is a good thing, E is astronomically large for even moderate m and n. For $m = 1$ this reduces to: bit q of j equals a. The tensor E is moreover sparse and perfectly regular: each question-slice (q fixed) holds exactly N nonzero entries: every map answers exactly one a divided evenly over the A answer rows, N/A apiece. Within a slice the nonzero row follows the odometer pattern of a numeral system: runs of A^q consecutive j share a digit, cycling through all A rows with period A^{q+1}.

The odometer pattern gives every entry of M the same compact closed form: for $m = 1$, bit q of j equals 0 exactly when $j \bmod 2^{q+1} < 2^q$, and equals 1 otherwise.

EXAMPLE

Each entry of M collects the eight maps of Table 4 whose truth table carries the right digit. Filling the entries in — note the period halving column by column, with the outer columns written in their simplified forms, $j \bmod 16 < 8$ being simply $j < 8$ and $j \bmod 2 < 1$ simply "j even" — and then substituting the values of p_j:

$$
\begin{aligned}
M \;&=\; \begin{array}{c|cccc}
\mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\
\hline
\mathbf{0} & \sum_{j<8} p_j & \sum_{j \bmod 8<4} p_j & \sum_{j \bmod 4<2} p_j & \sum_{j \text{ even}} p_j \\
\mathbf{1} & \sum_{j\geq 8} p_j & \sum_{j \bmod 8\geq 4} p_j & \sum_{j \bmod 4\geq 2} p_j & \sum_{j \text{ odd}} p_j
\end{array} \\
&=\; \begin{array}{c|cccc}
\mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\
\hline
\mathbf{0} & 0.585 & 0.779 & 0.890 & 0.818 \\
\mathbf{1} & 0.415 & 0.221 & 0.110 & 0.182
\end{array} \,.
\end{aligned}
\tag{26}
$$

The update rule also carries over verbatim. When question q is asked and the answer $a = \psi(q)$ comes back, the process of elimination reads, in one formula,

$$
p'_j \;=\; \frac{\delta\big(\phi_j(q),\, a\big)\; p_j}{M_{a,q}} \;=\; \frac{E_{a,q,j}\; p_j}{M_{a,q}}. \tag{27}
$$

The Kronecker delta performs the elimination. Only maps that agree with a survive and the division renormalizes the survivors, whose total prior mass is exactly $M_{a,q}$ by equation (24). Note that one and the same number prices the answer and funds the update: the event arrives with the probability $M_{a,q}$, costs Werner $-\log_2 M_{a,q}$ bits of surprisal, and its belief mass becomes the new unit of normalization. Every update shrinks the support by A^{-1}, radically thinning, especially for large m. The survivors live on a block-wave in j-space. The map determines the offset and frequency. The total set of survivors depends on the set of asked questions, not the order: updating commutes. This is illustrated in the example below.

EXAMPLE

Figure 3 makes the process visual for $n = m = 2$: a schematic prior over $N = 4^4 = 256$ maps, drawn as vertical lines with thickness representing weight. Each answer stops 3 out of every 4 surviving lines: maps whose truth table carry a wrong digit at the asked question. Renormalization inflates the weight of the rest, so the total probability is always unit. Four questions cut $256 \to 64 \to 16 \to 4 \to 1$, and the last line standing is the truth. Note the regularity of the cull: each answer keeps a single residue class of j, so the cadence of the survivors is regular.

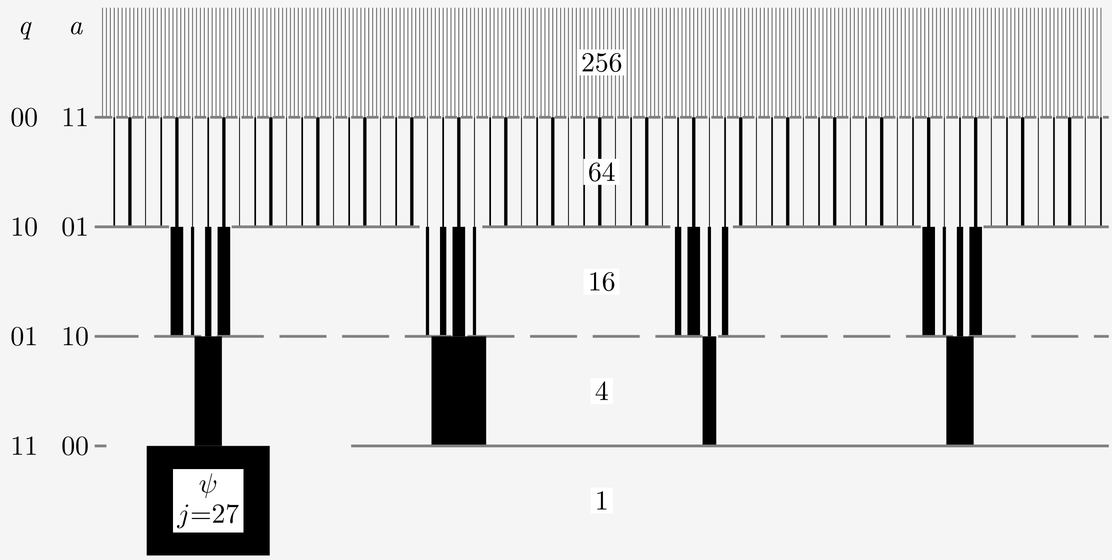


Figure 3: The update rule as a cull, for $n = m = 2$. All 256 candidate maps start as vertical lines under a schematic prior. The truth is the bitwise negation $\psi(q) = \neg q$ ($j = 27$), and the four questions are asked in the order $00, 10, 01, 11$. Each answer terminates the 3/4 of surviving maps whose truth-table digit at the asked question disagrees. Every survivor's thickness, proportional to its weight, is rescaled by the same renormalization factor, preserving their ratios. After four questions a single map remains. Interestingly, the first cull's survivors are evenly spaced because it's the smallest question. After the three lowest q, even spacing returns.

The total entropy of the table is extensive over the columns, exactly as in equation (12):

$$H(M) \;=\; \sum_{q\in\mathcal{Q}} H\big(M_{\cdot,q}\big) \;=\; -\sum_{q\in\mathcal{Q}}\sum_{a\in\mathcal{A}} M_{a,q}\log_2 M_{a,q} \;\leq\; mQ, \tag{28}$$

each column carrying at most m bits (a uniform spread over its A answers). The ceiling mQ is the size of the entire truth table in bits, and it is attained precisely by the uniform prior: maximal ignorance prices every answer at full cost.

The belief itself carries an entropy as well, and we give its gap to the table a name:

$$H(p) \;=\; -\sum_{j=0}^{N-1} p_j \log_2 p_j, \qquad C \;:=\; H(M) - H(p) \;\geq\; 0. \tag{29}$$

$H(p)$, Werner's uncertainty about the *identity* of the truth, obeys the same ceiling: a distribution over N maps has entropy at most $\log_2 N = mQ$, attained again by the uniform prior. The information in the prior is not directly useful for prediction, but it is good to track.

Why is $C > 0$? The answers jointly determine the map, so $H(p)$ is the *joint* entropy of all Q answers, while $H(M)$ sums their *marginal* entropies and a joint entropy never exceeds the sum of its marginals. Hence $H(p) \leq H(M)$, and the gap C, the **total correlation** of the answers, is nonnegative: knowledge stored in correlations *between* answers, invisible to every single column of M.

We choose this convention for $H(M)$ because in the ignorant case

1. $H(M)$ is the total amount of information we must collect from questions to settle the map

2. $H(M) = H(p)$, so the ignorance agrees in both bases, without correlations.

Therefore, the scaling is sensible. How these quantities respond to an intelligent prior, is the main question of this work.

Let us watch these ledgers in motion by playing the guiding example through all four questions. Werner can compute two forecasts for every unasked question, in advance, from M alone: the expected surprisal $\langle s \rangle$, which by equation (13) is the entropy of that question's column, and the expected drop $\langle \Delta H(M) \rangle$ of the *whole* table if that question is asked next. This is the average over possible a, weighted by their probability (according to M):

$$\langle s \rangle(q) \;=\; \sum_a M_{a,q}\left(-\log_2 M_{a,q}\right), \tag{30}$$

$$\langle \Delta H(M) \rangle(q) \;=\; \sum_a M_{a,q}\left[H(M) - H\left(M^{(q,a)}\right)\right], \tag{31}$$

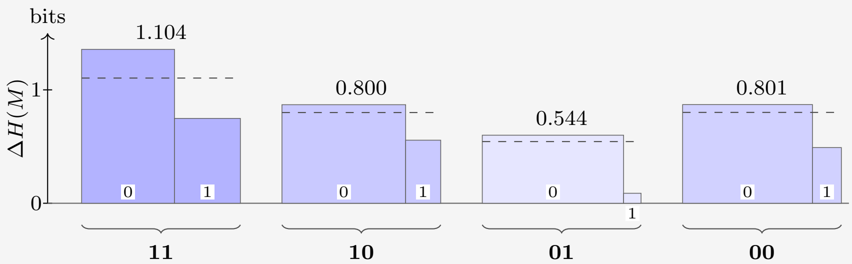
where $M^{(q,a)}$ is the table after the update rule (27) processes answer a to question q. We know the column of q will drop $\langle s \rangle(q)$ in entropy. But other columns will also respond. The excess of the second over the first is the expected transfer to unasked columns, and it can only be paid out of correlations. We append both forecasts as extra rows under M.

EXAMPLE

Werner plays greedily: always asking the question with the largest forecast. The truth is AND ($\psi = \phi_8$).

Question 1. The starting state:

a\q	**11**	**10**	**01**	**00**
0	0.585	0.779	0.890	0.818
1	0.415	0.221	0.110	0.182
$\langle s \rangle$	0.979	0.762	0.500	0.684
$\langle \Delta H(M) \rangle$	1.104	0.800	0.544	0.801



To see where the forecast entries come from, take $q = 11$: the answer $a = 0$ arrives with probability $M_{0,11} = 0.585$ and would drop $H(M)$ from 2.925 to 1.569; the answer $a = 1$ arrives with probability 0.415 and drops it only to 2.178, so

$$\langle s \rangle(11) = 0.585\,(0.773) + 0.415\,(1.269) = 0.979,$$
$$\langle \Delta H(M) \rangle(11) = 0.585\,(1.357) + 0.415\,(0.748) = 1.104.$$

Both forecast rows point to $q = 11$: it is the closest thing to a coin on offer ($\langle s \rangle = 0.979$) and also moves the whole table most, by a wide margin. Werner asks it, and nature answers $a = 1$ (probability 0.415, surprisal $s_1 = 1.269$). Eight maps die:

j	$\phi_j(11)$	name	p_j	p'_j
~~0~~	~~0~~	~~FALSE (constant 0)~~	~~0.394~~	~~—~~
~~1~~	~~0~~	~~$\neg(b_1 \lor b_0)$ (NOR)~~	~~0.010~~	~~—~~
~~2~~	~~0~~	~~$\neg b_1 \land b_0$~~	~~0.010~~	~~—~~
~~3~~	~~0~~	~~$\neg b_1$~~	~~0.010~~	~~—~~
~~4~~	~~0~~	~~$b_1 \land \neg b_0$~~	~~0.131~~	~~—~~
~~5~~	~~0~~	~~$\neg b_0$~~	~~0.010~~	~~—~~
~~6~~	~~0~~	~~$b_1 \oplus b_0$ (XOR)~~	~~0.010~~	~~—~~
~~7~~	~~0~~	~~$\neg(b_1 \land b_0)$ (NAND)~~	~~0.010~~	~~—~~
8	1	$b_1 \land b_0$ (AND)	0.213	0.513
9	1	$b_1 = b_0$ (XNOR)	0.092	0.222
10	1	b_0 (projection)	0.040	0.096
11	1	$b_1 \to b_0$ (IF THEN)	0.010	0.024
12	1	b_1 (projection)	0.010	0.024
13	1	$b_0 \to b_1$ (IF THEN)	0.030	0.072
14	1	$b_1 \lor b_0$ (OR)	0.010	0.024
15	1	TRUE (constant 1)	0.010	0.024

The ledger:

$$H(M)\colon 2.925 \to 2.178, \qquad H(p)\colon 2.707 \to 2.093, \qquad C\colon 0.218 \to 0.085.$$

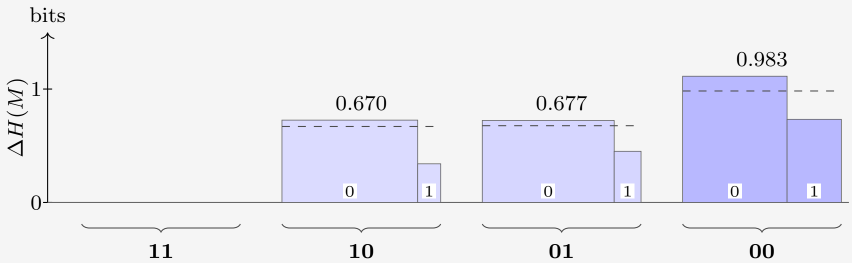
Werner received 1.269 bits but destroyed only 0.748 bits of table. Correlations point the right way only on average, not on every draw. In this case, Werner got the much less informative branch, and paid the premium price for it. The destroyed-per-received ratio of this round is 0.589: well below 1, anti-leverage.

Question 1 is a reminder that the forecast rows are *expectations*: they hold on average over many repetitions of the experiment, and only if the world really does draw its truth ψ from Werner's prior. On any single run the realized numbers are free to deviate.

EXAMPLE

Question 2. The state after the collapse, with fresh forecasts:

a\q	**11**	**10**	**01**	**00**
0	0	0.855	0.831	0.658
1	1	0.145	0.169	0.342
$\langle s\rangle$	—	0.596	0.655	0.927
$\langle \Delta H(M)\rangle$	—	0.670	0.677	0.983



Greedy picks $q = 00$. Nature answers $a = 0$ — the answer Werner favored (probability

0.658), so the surprisal is only $s_2 = 0.604$ bits. Four of the eight die:

j	$\phi_j(00)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.513	0.780
~~9~~	~~1~~	~~$b_1 = b_0$ (XNOR)~~	~~0.222~~	~~—~~
10	0	b_0 (projection)	0.096	0.147
~~11~~	~~1~~	~~$b_1 \to b_0$ (IF THEN)~~	~~0.024~~	~~—~~
12	0	b_1 (projection)	0.024	0.037
~~13~~	~~1~~	~~$b_0 \to b_1$ (IF THEN)~~	~~0.072~~	~~—~~
14	0	$b_1 \vee b_0$ (OR)	0.024	0.037
~~15~~	~~1~~	~~TRUE (constant 1)~~	~~0.024~~	~~—~~

The ledger:

$$H(M)\colon 2.178 \to 1.065, \qquad H(p)\colon 2.093 \to 1.035, \qquad C\colon 0.085 \to 0.030.$$

This time 1.113 bits are destroyed for 0.604 received. Where did the extra 0.509 come from? Partly from the favored answer arriving — the belief dropped 1.058 against only 0.604 paid — and partly from the store: C fell by 0.055. Step ratio 1.842. Cumulatively, however, 1.860 destroyed per 1.873 received gives a leverage of 0.993 after 2 steps: still a hair below 1. Even a strong second round has not quite repaid the expensive shock of question 1.

Why track C, when its step-by-step changes visibly do not match the amounts deduced? Because on a single realized run the deduced part has *two* sources, and C is the only quantity that separates them. Writing Δ for the realized drops in one step,

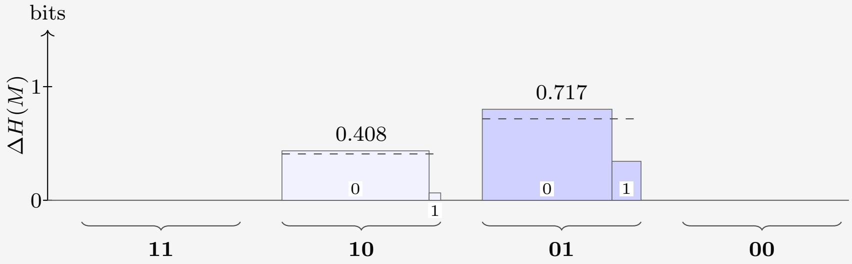
$$\underbrace{\Delta H(M) - s}_{\text{deduced}} = \underbrace{\left[\Delta H(p) - s\right]}_{\text{windfall}} + \underbrace{\left[-\Delta C\right]}_{\text{tower}}, \tag{32}$$

an identity, since $H(M) = H(p) + C$ at every stage. The windfall is luck: the realized surprisal undershooting (or overshooting) the realized drop in belief entropy: zero in expectation. The tower term is structure: a genuine withdrawal from the correlation store, and the channel with a global guarantee. Over a complete run its withdrawals sum to exactly the initial C, while the windfalls average away.

EXAMPLE

Question 3. Two questions remain:

a\q	**11**	**10**	**01**	**00**
0	0	0.927	0.817	1
1	1	0.073	0.183	0
$\langle s \rangle$	—	0.378	0.687	—
$\langle \Delta H(M) \rangle$	—	0.408	0.717	—



Greedy picks $q = 01$; nature answers $a = 0$ (probability 0.817, surprisal $s_3 = 0.292$). Two die:

j	$\phi_j(01)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.780	0.955
~~10~~	~~1~~	~~b_0 (projection)~~	~~0.147~~	—
12	0	b_1 (projection)	0.037	0.045
~~14~~	~~1~~	~~$b_1 \vee b_0$ (OR)~~	~~0.037~~	—

The ledger:

$$H(M)\colon 1.065 \to 0.264, \qquad H(p)\colon 1.035 \to 0.264, \qquad C\colon 0.030 \to 0.$$

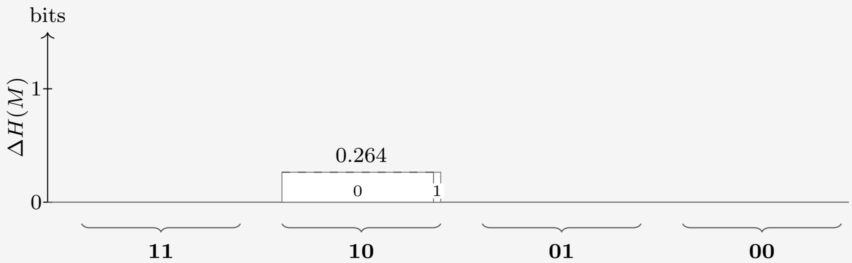
Again a bargain: 0.801 destroyed for a mere 0.292 received, and the cumulative ratio finally clears unity, at 1.229. And note that C has dropped to zero.

C measures correlations that are not visible from M. If there's only one unknown column, there's nothing to infer. Put another way, usually, the binary maps are an overcomplete basis for M, because their number $N \gg \dim(M) = 2^{m+n}$. That's not the case when the number of free parameters has been reduced by learning to 1 each.

EXAMPLE

Question 4. One question, one heavily loaded coin:

a\q	**11**	**10**	**01**	**00**
0	0	0.955	1	1
1	1	0.045	0	0
$\langle s \rangle$	—	0.264	—	—
$\langle \Delta H(M) \rangle$	—	0.264	—	—



The two forecasts now coincide: with $C = 0$ there is no transfer left to hope for, and the last question can only be observed, not leveraged. Nature answers $a = 0$ (probability 0.955, surprisal $s_4 = 0.066$):

j	$\phi_j(10)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.955	1.000
~~12~~	~~1~~	~~b_1 (projection)~~	~~0.045~~	~~—~~

The ledger closes:

$$H(M)\colon 0.264 \to 0, \qquad H(p)\colon 0.264 \to 0, \qquad C\colon 0 \to 0.$$

The truth is AND. The step destroyed 0.264 for 0.066 received (ratio 3.990) — not deduction this time, but luck: the destroyed amount was fixed in advance (the column's full entropy), while the realized surprisal undershot it because the likelier answer arrived. In the split of equation (32): pure windfall, tower 0.

Cumulatively: 2.925 bits of table destroyed, all of it, for 2.231 bits of surprisal received: a run ratio of 1.311. Most of the realized 1.311 was luck, not intelligence. Figure 4 stacks the books after each question: what remains, plus what was received. Once it dips below the starting entropy, our prior has done some work.

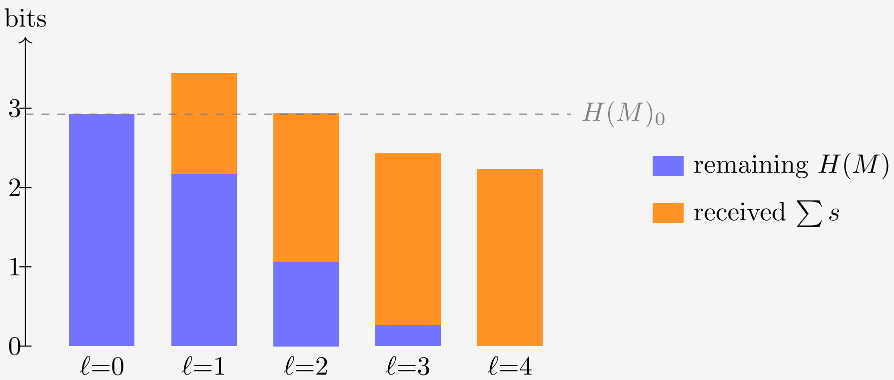


Figure 4: The realized run through the guiding example: remaining table entropy plus cumulative surprisal received, after each question, against the initial total (dashed). The gap between each stack and the line is the cumulative deduced part. At $\ell = 1$ the stack tops out 0.521 bits *above* the line — the negative deduction of question 1, plainly visible. At $\ell = 2$ the stack still grazes the line from above (0.013 bits); only from $\ell = 3$ on do the stacks fall short of it, as genuine deduction overtakes the early loss.

If the prior is good, meaning it actual models reality (the prevalence of maps), we expect graphs such as the one above to drop as questions come in. The ratio of table entropy destroyed per bit of surprisal received deserves a name.

The **leverage**. Run the protocol for k rounds: questions $q_1, \ldots, q_k$ are asked, the answers $a_i = \psi(q_i)$ are each processed by the update rule (27), producing the tables $M^{(i)}$, and round i costs the surprisal $s_i = -\log_2 M^{(i-1)}_{a_i, q_i}$. The cumulative leverage after k rounds is the table entropy destroyed per bit of surprisal received:

$$L_k \;:=\; \frac{H\bigl(M^{(0)}\bigr) - H\bigl(M^{(k)}\bigr)}{\sum_{i=1}^{k} s_i}\,. \tag{33}$$

A leverage of 1 is the uniform-prior baseline: every bit of table entropy destroyed was paid for at face value, one bit of surprisal each. Anything above 1 is deduction — table entropy removed without ever being observed.

4 Correlators

What kind of operator is the update rule (27)? In map space it is as simple as an operator gets: multiplication by a diagonal 0/1 mask, then renormalization. But there is a change of basis in which the same operation becomes a *convolution* with a two-term kernel, and in which the correlation store C stops being a single number and becomes visible coordinate by coordinate. The main advantage is that the effect of each update is well tabulated beforehand into a small number of targeted parameters.

This section builds that basis for $m = 1$; the final paragraph deals with general m.

First, re-encode each answer as a spin,

$$\sigma_q(\phi_j) \;:=\; (-1)^{\phi_j(q)} \;\in\; \{+1, -1\}, \tag{34}$$

so answer $0 \mapsto +1$ and $1 \mapsto -1$. A pure relabeling, but it buys the one algebraic fact everything below turns on: $\sigma_q^2 = 1$.

The **Walsh–Hadamard transform** is built from a single matrix. For one bit,

$$W \;=\; \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad W_{c,b} \;=\; (-1)^{c\,b}, \quad c, b \in \{0, 1\}. \tag{35}$$

EXAMPLE

Acting on a one-bit distribution,

$$W \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \;=\; \begin{pmatrix} p_0 + p_1 \\ p_0 - p_1 \end{pmatrix} \;=\; \begin{pmatrix} 1 \\ \langle\sigma\rangle \end{pmatrix} :$$

row $c = 0$ computes the normalization, row $c = 1$ the spin bias. Distribution in, correlators out.

The normalization riding along is a redundancy. The original distribution also uses one variable more than its degrees of freedom.

EXAMPLE

Tensoring two copies (the $n = 1$ case) does the same for two bits at once:

$$W \otimes W \;=\; \begin{pmatrix} W & W \\ W & -W \end{pmatrix} \;=\; \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{array}{l} \leftarrow c = 00\colon\ \text{sums to } 1 \\ \leftarrow c = 01\colon\ \langle\sigma_0\rangle \\ \leftarrow c = 10\colon\ \langle\sigma_1\rangle \\ \leftarrow c = 11\colon\ \langle\sigma_1\sigma_0\rangle \end{array} \tag{36}$$

The Kronecker product multiplies signs, so for any number l of tensor factors the entry at row c, column j is

$$\big(W^{\otimes l}\big)_{c,j} \;=\; \prod_{b=0}^{l-1} (-1)^{c_b\, j_b} \;=\; (-1)^{\sum_b c_b\, j_b}, \tag{37}$$

with c_b, j_b the b-th bits of the row and column indices; every further tensor factor appends one more bit.

For $m = 1$ a map *is* its truth table of Q bits — the numeral j — so Werner's belief is a vector of length $N = 2^Q$, and one transform computes every correlator at once:

$$\chi \;=\; W^{\otimes Q} p, \qquad \chi_{\mathcal{S}} \;=\; \Big\langle \prod_{q\in\mathcal{S}} \sigma_q \Big\rangle \;=\; \sum_{j=0}^{N-1} (-1)^{\sum_{q\in\mathcal{S}} \phi_j(q)}\, p_j. \tag{38}$$

Each row computes the **correlator** of one subset $\mathcal{S} \subseteq \mathcal{Q}$ of questions. In the subscript we may encode the set as its *mask*: written in the same descending question order as everywhere else, bit q of the subscript records whether $q \in \mathcal{S}$. The row index of the matrix, read as a numeral, is the mask.

EXAMPLE

For $n = 2$, the question order is $(11, 10, 01, 00)$ so χ_{0110} is the correlator of $\mathcal{S} = \{10, 01\}$.

the sixteen rows of $W^{\otimes 4}$ probe the sixteen subsets of $Q = \{11, 10, 01, 00\}$. A few entries of the dictionary:

mask	set $\mathcal{S}$	correlator
0000	$\emptyset$	$\chi_{0000} = 1$ (normalization)
0001	$\{00\}$	$\chi_{0001} = \langle \sigma_{00} \rangle$
1000	$\{11\}$	$\chi_{1000} = \langle \sigma_{11} \rangle$
0110	$\{10, 01\}$	$\chi_{0110} = \langle \sigma_{10}\, \sigma_{01} \rangle$
1111	$\{11, 10, 01, 00\}$	$\chi_{1111} = \langle \sigma_{11}\, \sigma_{10}\, \sigma_{01}\, \sigma_{00} \rangle$, the full parity

Sort the N correlators by $|\mathcal{S}|$, the number of questions probed, into *shells*. Shell 0 is the normalization, $\chi_\emptyset = 1$, riding along as a correlator. Shell 1 is M in different clothes:

$$\chi_{\{q\}} = M_{0,q} - M_{1,q}, \qquad M_{a,q} = \tfrac{1}{2}\big(1 + (-1)^a\, \chi_{\{q\}}\big), \tag{39}$$

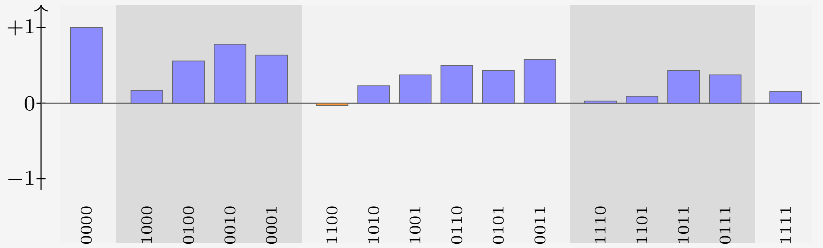
the bias of column q. For $m = 1$, we elegantly see the first shell correlators are the distance from fair coins.

Shell 2 holds the agreement biases of pairs of questions — knowledge M cannot see: two priors can share every column of M and differ here. Higher shells store subtler and subtler parity knowledge. Nothing is lost in the sorting: $W^2 = 2I$, so $(W^{\otimes Q})^2 = NI$ — the transform is its own inverse up to a factor N, and knowing every correlator is knowing the belief. The ledger also gives C a signature: a product prior ($C = 0$) factorizes every correlator, $\chi_{\mathcal{S}} = \prod_{q \in \mathcal{S}} \chi_{\{q\}}$, so its upper shells repeat what shell 1 already said; correlated knowledge is precisely *non-factorizing* correlators.

EXAMPLE

The prior of Table 4 transforms to the ledger

shell	correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = +0.170,\ \ \chi_{0100} = +0.558,\ \ \chi_{0010} = +0.780,\ \ \chi_{0001} = +0.636$
2	$\chi_{1100} = -0.032,\ \ \chi_{1010} = +0.230,\ \ \chi_{1001} = +0.374,$
	$\chi_{0110} = +0.498,\ \ \chi_{0101} = +0.434,\ \ \chi_{0011} = +0.576$
3	$\chi_{1110} = +0.028,\ \ \chi_{1101} = +0.092,\ \ \chi_{1011} = +0.434,\ \ \chi_{0111} = +0.374$
4	$\chi_{1111} = +0.152$



Shell 1 reproduces the columns of equation (26): $(1+0.170)/2 = 0.585 = M_{0,11}$, and likewise 0.779, 0.890, 0.818. The negative entry χ_{1100} says questions 11 and 10 disagree slightly more often than they agree; every other pair leans toward agreement. This is a reflection of the strong coherent 0-preference encoded in the prior. Answering 1 would give negative 1-shells.

Now look at the *rows* of $W^{\otimes Q}$ as functions of j. They are **block waves**. The observation mask is built of the same material:

$$\delta\big(\phi_j(q), a\big) \;=\; \frac{1}{2}\Big(1 + (-1)^{a+\phi_j(q)}\Big), \tag{40}$$

a 0/1 block wave — one constant plus one Hadamard row. And multiplication and convolution swap under this transform exactly as under Fourier: multiplying two functions of j pointwise convolves their ledgers. And the observation mask's ledger has only two nonzero entries — at $\emptyset$ and at $\{q\}$ — so this convolution is no great sum: it combines each correlator with exactly one other.

The update rule, carried out entirely in the ledger, is therefore that two-term combination, followed by renormalization by the new shell 0. In index notation, learning $a = \psi(q)$ sends every set $\mathcal{S}$ to

$$\chi'_{\mathcal{S}} \;=\; \frac{\chi_{\mathcal{S}} + (-1)^a\,\chi_{\mathcal{S}\triangle\{q\}}}{1 + (-1)^a\,\chi_{\{q\}}}, \tag{41}$$

where $\mathcal{S}\triangle\{q\}$ is $\mathcal{S}$ with the membership of q toggled: q is added if absent, removed if present (on masks, bit q is flipped). The numerator is the monomial algebra of $\sigma_q^2 = 1$: multiplying $\prod_{r\in S}\sigma_r$ by σ_q toggles q's membership of the product, so every correlator is averaged with its **partner**: the correlator differing by q, signed by the answer. The denominator is the $\mathcal{S} = \emptyset$ instance of the numerator, and by equation (39) it equals $2M_{a,q}$: the same number that priced the answer in equation (27) renormalizes the ledger.

EXAMPLE

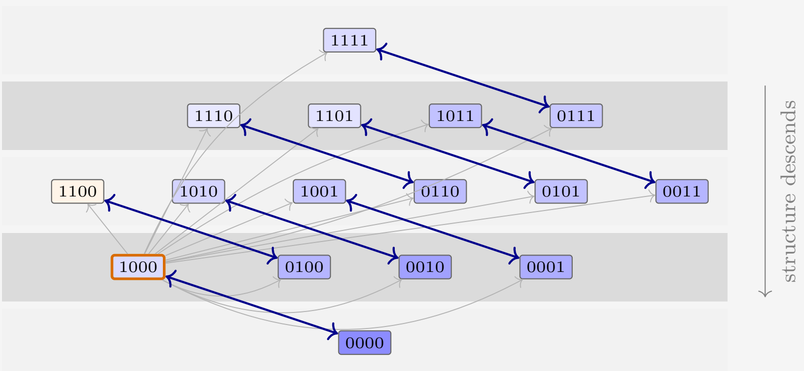


Figure 5: The update rule in the ledger, drawn for the guiding example's first question ($q = 11$). Every set is paired with the partner whose membership of 11 is toggled (arrows); each pair is averaged with the answer's sign and renormalized, per equation (41). The pair $\emptyset \leftrightarrow \{11\}$ at the orange-bordered asked set fixes the normalization and saturates the asked bias; each blue double arrow averages a pair into both of its members, and the net flow of structure is one shell down, toward M. The thin gray arrows from the asked q to every other pill remind that this sets the renormalization. The fill encodes each prior value to scale: darker is larger, orange is negative.

Read formula (41) by cases:

- $\mathcal{S} = \emptyset$: partner $\{q\}$, giving $(1 + (-1)^a \chi_{\{q\}})/(1 + (-1)^a \chi_{\{q\}}) = 1$. Normalization is preserved.

- $\mathcal{S} = \{q\}$, the asked question: partner $\emptyset$, giving $(\chi_{\{q\}} + (-1)^a)/(1 + (-1)^a \chi_{\{q\}}) = (-1)^a$. The bias saturates to certainty — the one-hot collapse of column q, derived rather than decreed.

- $\mathcal{S} = \{q'\}$, an unasked question: the partner is the pair $\{q', q\}$, and one line of algebra puts the shift in terms of the *connected* correlator:

$$\chi'_{\{q'\}} - \chi_{\{q'\}} = \frac{(-1)^a \big(\chi_{\{q',q\}} - \chi_{\{q'\}}\, \chi_{\{q\}}\big)}{2M_{a,q}} = \frac{(-1)^a \operatorname{Cov}\big(\sigma_{q'}, \sigma_q\big)}{2M_{a,q}}. \tag{42}$$

  Unasked columns move in proportion to their covariance with the asked question: knowledge stored one shell up descends into the predictions. A product prior has zero covariance everywhere, so its unasked columns are frozen — the "$C = 0$ means no transfer" fact, now a one-line consequence.

- Higher shells: the same cascade. Each update marries shell-k sets containing q to shell-($k-1$) sets without it; whatever structure was stored jointly with q moves one shell down, toward M.

EXAMPLE

Question 1 of the walkthrough asked $q = 11$ and received $a = 1$: sign $(-1)^a = -1$, toggled set $\{11\}$, mask 1000. Equation (41) maps the ledger of the previous box to

shell	updated correlators
0	$\chi'_{0000} = 1$
1	$\chi'_{1000} = -\mathbf{1}$, $\chi'_{0100} = +0.711$, $\chi'_{0010} = +0.663$, $\chi'_{0001} = +0.316$
2	$\chi'_{1100} = -0.711$, $\chi'_{1010} = -0.663$, $\chi'_{1001} = -0.316$,
	$\chi'_{0110} = +0.566$, $\chi'_{0101} = +0.412$, $\chi'_{0011} = +0.171$
3	$\chi'_{1110} = -0.566$, $\chi'_{1101} = -0.412$, $\chi'_{1011} = -0.171$, $\chi'_{0111} = +0.267$
4	$\chi'_{1111} = -0.267$

The asked bias saturates: $\chi'_{1000} = (0.170 - 1)/(1 - 0.170) = -1$. The other singletons move by equation (42): for $q' = 10$ the covariance is $-0.032 - (0.558)(0.170) = -0.127$, so the shift is $(-1)(-0.127)/(2 \times 0.415) = +0.153$, landing on $+0.711$ — and indeed $(1 + 0.711)/2 = 0.855$, the column $M_{0,10}$ of the Question 2 box. Note also that every mask containing the asked bit is now just $(-1)^a$ times its partner ($\chi'_{1010} = -\chi'_{0010}$, etc.): with σ_{11} pinned, no correlator involving question 11 carries independent knowledge anymore.

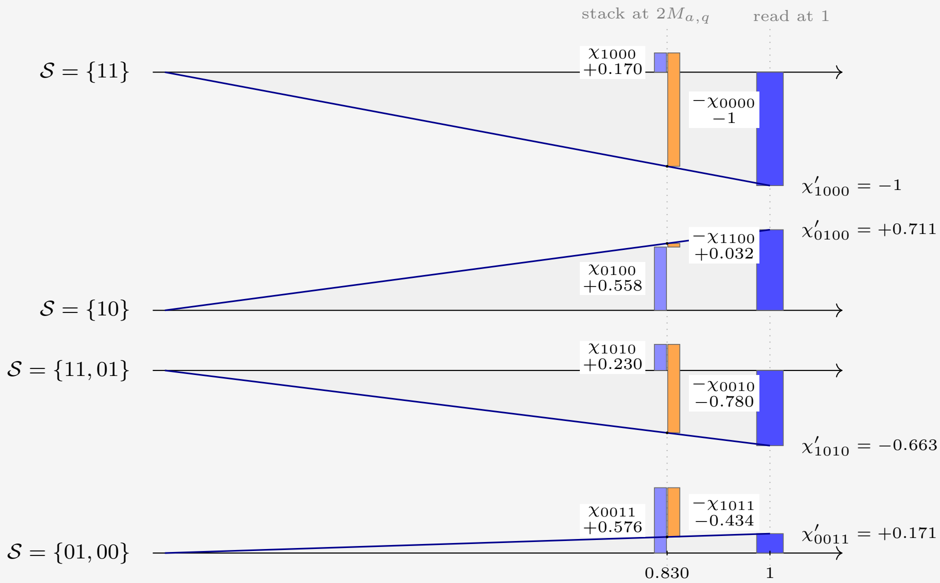


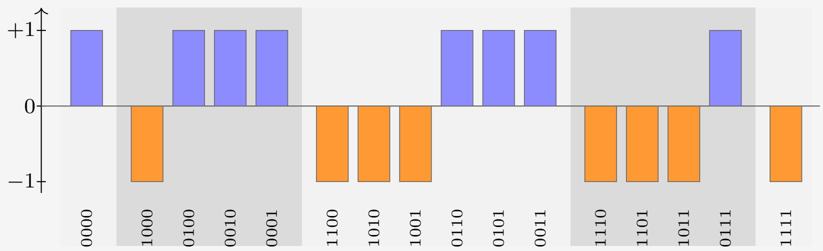
Figure 6: The update rule of equation (41), drawn for the guiding example's first update ($q = 11$, $a = 1$, so $(-1)^a = -1$ and toggled bit 1000). One row per set $\mathcal{S}$. At $x = 2M_{a,q} = 1 + (-1)^a \chi_{1000} = 0.830$ stands the numerator as a waterfall: the bar of $\chi_{\mathcal{S}}$ (blue), and beside it the partner's signed contribution $(-1)^a \chi_{\mathcal{S} \triangle \{q\}}$ (orange), hanging from the level where the first bar ends. The two subscripts differ by the flipped bit. The ray from the origin through the waterfall's endpoint rescales it, by similar triangles, to $x = 1$, where its height is the updated correlator $\chi'_{\mathcal{S}}$.

As Werner moves through the questions, more correlators settle to ± 1. The 1-shell models the belief of the bias of each question. The 2-shell answers questions such as 'If the answer to $\psi(10) = 0$, then $\psi(01) = 1$'. And the higher orders are even more conditional, measuring belief about such statements, in the case of certain answers, etc. As we learn more, the conditions become satisfied or not, and the information becomes less nested.

EXAMPLE

The remaining questions of the run (00, 01, 10, all answered 0, so with sign +1) each saturate their own bit the same way, and since XOR toggles commute, the order does not matter — the commutativity of updating, seen again. After all four, the ledger is pure signs, $\chi_{\mathcal{S}} = (-1)^{|\mathcal{S}\cap\{11\}|}$ — -1 on every set containing 11, $+1$ on the rest: the transform of the point mass on $\psi = \phi_8$:

shell	final correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = -1,\ \chi_{0100} = +1,\ \chi_{0010} = +1,\ \chi_{0001} = +1$
2	$\chi_{1100} = -1,\ \chi_{1010} = -1,\ \chi_{1001} = -1,$
	$\chi_{0110} = +1,\ \chi_{0101} = +1,\ \chi_{0011} = +1$
3	$\chi_{1110} = -1,\ \chi_{1101} = -1,\ \chi_{1011} = -1,\ \chi_{0111} = +1$
4	$\chi_{1111} = -1$



4.1 General m

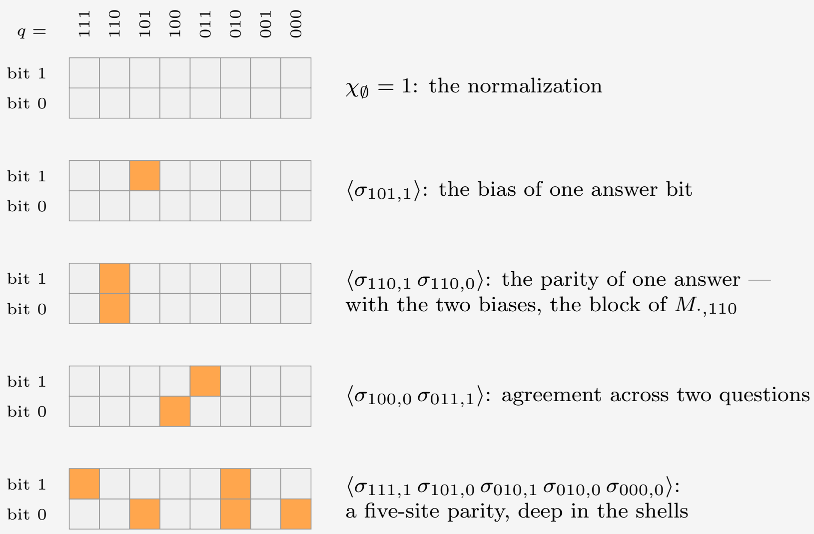
It turns out that taking $m > 1$ is more general but not much more instructive. With multiple output bits it is as if we have m a priori independent maps per question; they just happen always to be asked and answered simultaneously, in unison. Therefore we do not have to think about what one of them teaches us about another. It would mean we have more than 2 options every question, and therefore the factor by which every update thins the hypotheses is not 2 but A. The guiding example helps because the maps are represented as binary numbers, which are more familiar.

For completeness, the main formulas. The truth table is now mQ bits: one *site* (q,i) per bit i of each answer, each carrying a spin $\sigma_{q,i}(\phi_j) := (-1)^{\phi_j(q)_i}$, where $\phi_j(q)_i$ is the i-th bit of the answer. Correlators are indexed by sets $\mathcal{S}$ of sites — masks of mQ bits under the same numeral convention, m of them per question:

$$\chi_{\mathcal{S}} = \Big\langle \prod_{(q,i)\in\mathcal{S}} \sigma_{q,i} \Big\rangle = \sum_{j=0}^{N-1} (-1)^{\sum_{(q,i)\in\mathcal{S}} \phi_j(q)_i}\, p_j. \tag{43}$$

EXAMPLE

For $n = 3$, $m = 2$ the sites form a 2×8 grid: one column per question, one row per bit of the answer. A set $\mathcal{S}$ is a pattern of hot cells in this grid, and $\chi_{\mathcal{S}}$ multiplies the corresponding spins:



Any of the $2^{16} = 65536$ hot/cold patterns is a correlator, and sorting by the number of hot cells gives the shells. The mask of $\mathcal{S}$ is the grid read as a 16-bit numeral.

Write $\mathcal{B}_q$ for the block of q's m sites, and $a \cdot \mathcal{T} := \sum_{i \in \mathcal{T}} a_i$ for the parity of the answer a on a subset $\mathcal{T} \subseteq \mathcal{B}_q$. The answer matrix is read off the correlators supported on one block — A of them per column:

$$M_{a,q} \;=\; \frac{1}{A} \sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{T}}. \tag{44}$$

The observation mask at (q, a) is a function of the m bits of one block, so its ledger has A nonzero entries, and hearing $a = \psi(q)$ combines every correlator with its A partners — one per toggle $\mathcal{T}$ within the asked block:

$$\chi'_{\mathcal{S}} \;=\; \frac{\sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{S} \triangle \mathcal{T}}}{A\, M_{a,q}}, \tag{45}$$

the denominator once more the $\mathcal{S} = \emptyset$ instance of the numerator. For $m = 1$ the three formulas collapse to equations (38), (39) and (41).

EXAMPLE

To draw the update, take a sparse prior at $n = 3$, $m = 2$: four maps, $\phi(q) = q \bmod 4$ (weight 0.45), constant 00 (0.35), "11 iff $q \geq 4$" (0.12), constant 11 (0.08). Werner asks $q = 110$ and hears $a = 11$: toggle signs $(+, -, -, +)$ on $(\emptyset, 110_0, 110_1, \text{both})$, where q_i denotes site (q, i), and denominator $AM_{11,110} = 0.8$. The pairs of the $m = 1$ figure become *quartets*:

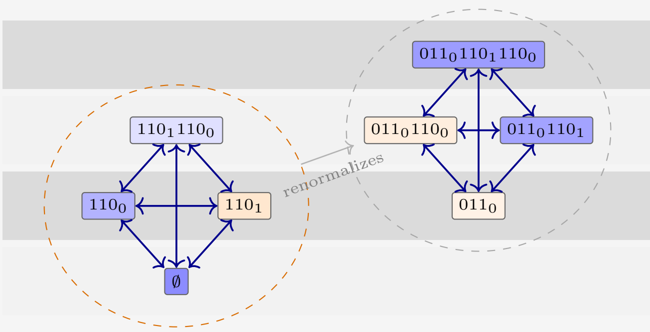


Figure 7: For $m = 2$, toggling any subset of the asked block's two sites ties four correlators into a clique that averages jointly, per equation (45). The on-block quartet (orange hull) contains the normalization and the entire block of $M_{\cdot,110}$; its signed sum is the denominator that renormalizes every other quartet. Fills to scale for the sparse prior, as before.

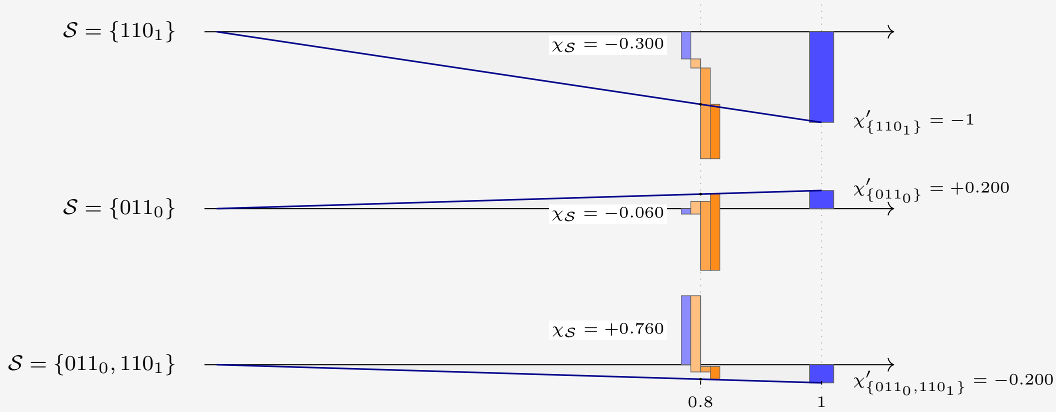


Figure 8: The $m = 2$ update, three sets from the sparse prior. The waterfall now has four segments: $\chi_{\mathcal{S}}$ (blue) and its three signed quartet partners, in toggle order 110_0, 110_1, both (light to dark orange). The ray from the origin rescales the endpoint from $x = AM_{a,q} = 0.8$ to $x = 1$. The asked site saturates to -1; the foreign site 011_0 flips sign, $-0.060 \to +0.200$; and $\{011_0, 110_1\}$ lands on -0.200, the slaved value $-\chi'_{\{011_0\}}$.

5 Expectations

In expectation over the truth $\psi \sim p$, as well as over the question order, we take the ratio of the expected totals,

$$\langle L_\ell \rangle \;:=\; \frac{H\big(M^{(0)}\big) - \big\langle H\big(M^{(\ell)}\big)\big\rangle}{\big\langle \sum_{i=1}^{\ell} s_i \big\rangle} \,. \tag{46}$$

We did this implicitly in Section 2.1. There, the problem was actually symmetric in $q_0 \leftrightarrow q_1$. That is not the case in general.

In order to progress, we will be interested in such averages. Both are natural. The question order is uniform, without replacement: before any evidence arrives, there is no reason to prefer one question over another. And the truth is drawn from the prior itself: Werner scores his future against his own beliefs, and he has faith in them.

The bookkeeping simplifies at once. Updating commutes, so after ℓ rounds the state depends only on the asked *set* $\mathcal{S}$, a uniform random subset of size ℓ. And the received surprisals telescope to the surprisal of the whole answer block,

$$\sum_{i=1}^{\ell} s_i \;=\; -\log_2 \Pr\big(\psi(\mathcal{S}) = \psi(\mathcal{S})\big), \tag{47}$$

whose average over ψ is a block entropy. Everything is therefore generated by one sequence, the mean block entropy at size k:

$$G_k \;:=\; \binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} H(\psi(\mathcal{S})), \qquad G_0 = 0, \qquad G_Q = H(p). \tag{48}$$

G_1 is the mean column entropy, so $H(M)_0 = QG_1$. Two exact laws follow. The first is equation (47) averaged:

$$\Big\langle \sum_{i=1}^{\ell} s_i \Big\rangle \;=\; G_\ell. \tag{49}$$

For the second, fix $\mathcal{S}$ and average the posterior table over the answers. Each unasked column's entropy becomes a conditional entropy, $H(\psi(q') \mid \psi(\mathcal{S})) = H(\psi(\mathcal{S} \cup \{q'\})) - H(\psi(\mathcal{S}))$, and the asked columns contribute zero. Now average over $\mathcal{S}$: every set of size $\ell + 1$ arises as $\mathcal{S} \cup \{q'\}$ once per element, and counting with $(\ell{+}1)\binom{Q}{\ell+1} = (Q{-}\ell)\binom{Q}{\ell}$,

$$\big\langle H\big(M^{(\ell)}\big)\big\rangle \;=\; (Q - \ell)\,\big(G_{\ell+1} - G_\ell\big). \tag{50}$$

The whole expected learning curve is the discrete derivative of one monotone sequence. By conservation, destruction and deduction are the differences $H(M)_0 - \langle H(M^{(\ell)})\rangle$ and $H(M)_0 - \langle H(M^{(\ell)})\rangle - G_\ell$, and equation (46) becomes

$$\langle L_\ell\rangle \;=\; \frac{H(M)_0 - (Q - \ell)\big(G_{\ell+1} - G_\ell\big)}{G_\ell}\,. \tag{51}$$

In terms of the prior, the block entropies are explicit. The maps that answer the pattern b on the asked set $\mathcal{S}$ form a residue class, with mass

$$G_\ell \;=\; -\binom{Q}{\ell}^{-1} \sum_{|\mathcal{S}|=\ell}\,\sum_{b} P_{S,b}\log_2 P_{S,b}, \qquad P_{S,b} \;:=\; \sum_{j\,:\,\phi_j(q)=b_q\ \forall q\in\mathcal{S}} p_j, \tag{52}$$

the inner sum running over the 2^ℓ answer patterns b on $\mathcal{S}$. Nothing here is specific to $m = 1$: blocks of m-bit answers obey the same laws verbatim, with b running over the $(A)^\ell$ patterns of the block.

Some observations.

- Expectation kills the windfall. Each step has $\langle \Delta H(p) - s\rangle = 0$, so the expected deduced part is a pure tower withdrawal: $H(M)_0 - \langle H(M^{(\ell)})\rangle - G_\ell = C_0 - \langle C_\ell\rangle$, with $\langle C_\ell\rangle = \langle H(M^{(\ell)})\rangle - \big(H(p) - G_\ell\big)$. It is nonnegative: anti-leverage is only ever bad luck.
- Deduction saturates one round early: substituting $G_Q = H(p)$ at $\ell = Q - 1$ already gives $\langle C_{Q-1}\rangle = 0$. The store is empty before the last question, which can only be observed, never leveraged.
- $\langle L_\ell\rangle \geq 1$ always, but it need not be monotone. The endpoint is forced: $\langle L_Q\rangle = H(M)_0/H(p) = 1 + C_0/H(p)$, the run ratio quoted in Section 3. This is also why equation (46) is a ratio of expectations: the expectation of the ratio is dominated by lucky runs with vanishing denominators.
- Sanity: $\ell = 0$ returns $H(M)_0 = QG_1$, and at $n = 1$ the laws reproduce the one-parameter world of Section 2.1.

There is also a monotonicity theorem hiding in G. Entropy is submodular, so the increments $G_\ell - G_{\ell-1}$, which are the expected surprisals of the ℓ-th answer, decrease with ℓ: answers get cheaper as evidence accumulates. Greedy play does the opposite on purpose, seeking out the most expensive question on the board.

EXAMPLE

For the guiding prior the block-entropy sequence is $G_k = (0,\ 0.731,\ 1.436,\ 2.100,\ 2.707)$, and the laws unfold it into the expected run:

ℓ	received $\langle \sum_i s_i \rangle$	remaining $\langle H(M^{(\ell)}) \rangle$	store $\langle C_\ell \rangle$	leverage $\langle L_\ell \rangle$
1	0.731	2.113	0.137	1.110
2	1.436	1.328	0.056	1.113
3	2.100	0.608	0	1.104
4	2.707	0	0	1.081

The expected leverage humps at $\ell = 2$ and lands on $2.925/2.707 = 1.081$. The realized run of Section 3 scattered around this gentle expectation: greedy questioning and a lucky truth delivered 1.311. Figure 9 stacks the expected books.

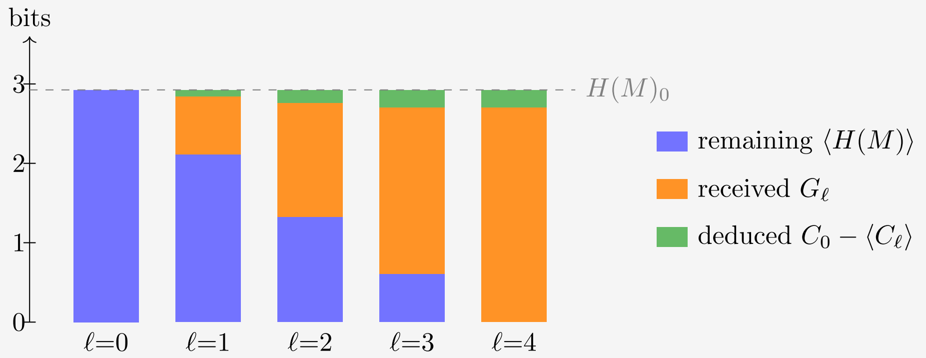


Figure 9: The expected run through the guiding example. Conservation is now exact at every ℓ: remaining plus received plus deduced reaches $H(M)_0$, so the cake always tops out on the dashed line, and no overshoot is possible. The top layer is the expected withdrawal from the correlation store, $C_0 - \langle C_\ell \rangle$; it saturates at $C_0 = 0.218$ already at $\ell = 3$.

5.1 Choosing the questions

The uniform order was a symmetry choice, not a strategy. Werner can do better along two independent axes: how far he looks ahead, and whether he commits to an order beforehand or lets each question depend on the answers so far. *Greedy* asks, at every step, the single question with the largest expected drop of $H(M)$: one step of lookahead. *ℓ-optimal* maximizes the expected knowledge after exactly ℓ questions, looking the whole horizon ahead. All four combinations are straightforward with the machinery at hand.

	order fixed beforehand	contingent on answers
greedy	Grow the asked set one question at a time, each maximizing the expected drop of $H(M)$ averaged over the still-unseen answers. One pass of the bookkeeping of equation (52); front-loads the cheap destruction, but cannot react to luck.	Recompute the forecasts of Section 3 after every answer and ask the current argmax: the walkthrough's play. One update per step; beats its fixed twin whenever an answer reshuffles the ranking.
ℓ-optimal	Updating commutes, so a fixed plan is just a set: search the $\binom{Q}{\ell}$ subsets for the smallest expected remainder $\sum_{q' \notin \mathcal{S}} H(\psi(q') \mid \psi(\mathcal{S}))$. The best non-adaptive knowledge; sees correlations that the one-step view misses.	Backward induction over posteriors, $V_t = \min_q \mathbb{E}_a V_{t-1}$ with $V_0 = H(M)$. The Bayes-optimal policy: dearest to compute, and the ceiling the other three approximate.

Expected knowledge improves left to right and top to bottom. The dynamic program in the last cell is exact but grows quickly: its states are the reachable posteriors, one per pair of asked set and answer pattern, $(1 + A)^Q$ in all. That is 81 for the guiding example and astronomical beyond; its policy is also horizon-dependent, since the best first question for $\ell = 1$ need not open the best plan for $\ell = 2$. For the guiding prior, contingent greedy already attains the contingent optimum at every horizon: expected remaining 1.821, 0.903, 0.334 bits, against the uniform 2.113, 1.328, 0.608. That is a kindness of this prior, not a law: a prior that stores its knowledge in deep shells can defeat greedy in either column, since a question worth little now may unlock a cascade later.

6 The Gibbs complexity prior

Every prior so far was engineered: a mechanism first, a distribution second. This section builds the prior the other way around, from a single principle – simple explanations are likelier – with circuit complexity as the measure of simplicity, made quantitative as a Gibbs ensemble over the hypothesis space.

6.1 NAND complexity, and two cheaper descriptors

Fix the universal gate: the two-input NAND. The complexity X_j of a map ϕ_j is the fewest gates in any feed-forward circuit that computes it. For the bookkeeping we count in the equivalent *and-inverter* form: two-input AND gates of unit cost, with inversion free on every wire – an edge attribute, not a gate. De Morgan's laws push inverters along wires and into gates, so an and-inverter circuit converts to a NAND-only circuit of the same gate count and back; X_j is the standard AIG-size metric of logic synthesis. Reference values: AND, OR, NAND and NOR all cost 1; XOR costs 3; constants, projections and negations cost 0. For $m > 1$ outputs, X_j is the size of one *joint* circuit computing all output bits together, gates shared – the honest minimum-hardware cost, smaller in general than the sum of per-output circuits.

The tabulation is feasible because X_j is invariant under relabeling: permuting or negating inputs, negating or (for $m = 2$) swapping outputs are all free operations, so the 65,536 maps of the $(4,1)$ and $(3,2)$ spaces collapse to 222 and 308 equivalence classes. The $(4,1)$ class values are published optimal AIG sizes[1]; the $(3,2)$ values come from that synthesizer: circuit sizes $r = 0, 1, 2, \ldots$ are encoded as satisfiability instances, and the first satisfiable r is provably minimal, every smaller size having been refuted.

Alongside the interior cost X_j, which takes a SAT solver to see, two descriptors are visible in one pass over the truth table:

- the *footprint* $F_j = nI_j + S_j$, where the support S_j counts the input bits the map actually depends on and the image size I_j counts the distinct answers it attains (from 1 for constants up to A). The combination is lossless – base-n digits, quotient I, remainder S – and reads as the map's *interface*: does it react to all its inputs, does it use all its values?
- the *bias* B_j: the number of 1s minus the number of 0s among the mQ truth-table bits. This is the symmetry breaker: X_j and F_j are even under negating all outputs, B_j is odd – an energy term coupling to it acts like a magnetic field, switching on the odd correlator shells that a pure complexity prior forbids.

For the guiding $(2,1)$ space:

j	map	X_j	S_j	I_j	F_j	B_j
0	FALSE	0	0	1	2	-4
1	$\neg(b_1 \vee b_0)$ (NOR)	1	2	2	6	-2
2	$\neg b_1 \wedge b_0$	1	2	2	6	-2
3	$\neg b_1$	0	1	2	5	0
4	$b_1 \wedge \neg b_0$	1	2	2	6	-2
5	$\neg b_0$	0	1	2	5	0
6	$b_1 \oplus b_0$ (XOR)	3	2	2	6	0
7	$\neg(b_1 \wedge b_0)$ (NAND)	1	2	2	6	$+2$
8	$b_1 \wedge b_0$ (AND)	1	2	2	6	-2
9	$b_1 = b_0$ (XNOR)	3	2	2	6	0
10	b_0 (projection)	0	1	2	5	0
11	$b_1 \to b_0$	1	2	2	6	$+2$
12	b_1 (projection)	0	1	2	5	0
13	$b_0 \to b_1$	1	2	2	6	$+2$
14	$b_1 \vee b_0$ (OR)	1	2	2	6	$+2$
15	TRUE	0	0	1	2	$+4$

Grouped by the full tuple (X, F, B), the sixteen maps fall into six energy classes:

(X, F, B)	members	(X, F, B)	members
$(0, 2, -4)$	1	$(1, 6, -2)$	4
$(0, 2, +4)$	1	$(1, 6, +2)$	4
$(0, 5, 0)$	4	$(3, 6, 0)$	2

At the larger sizes the tuple breakdown runs to hundreds of rows (102 distinct tuples at $(4,1)$, 196 at $(3,2)$), so we tabulate the complexity tiers alone:

[1] The `github.com/krinkin/bounds` reproducibility dataset, cross-checked against our own SAT-based synthesizer on random classes with zero mismatches.

X	$(2,1)$	$(4,1)$	$(3,2)$
0	6	10	64
1	8	48	432
2	—	256	2,064
3	2	940	4,860
4	—	2,336	10,832
5	—	6,464	16,208
6	—	10,616	17,060
7	—	18,984	10,672
8	—	17,680	3,312
9	—	7,882	32
10	—	320	—
total	16	65,536	65,536

The tiers are thin at both ends: ten gate-free maps at $(4,1)$ (two constants, eight literals) against only 320 at the ceiling $X = 10$. The typical function is middling-complex, and it is exactly this bulk that a complexity energy pushes against. Note also that $(2,1)$ skips $X = 2$ entirely: with two inputs, nothing a two-gate circuit computes is out of reach of one gate or zero.

6.2 The ensemble

Take the energy linear in the three descriptors and the prior as its Boltzmann weight,

$$E_j = \beta\, X_j + \alpha\, F_j + \mu\, B_j, \qquad p_j = \frac{e^{-E_j}}{\sum_k e^{-E_k}}, \tag{53}$$

with no overall temperature: the couplings are the fields. Positive β suppresses complex maps, positive α suppresses large footprints, and μ couples to the signed bias, $\mu < 0$ tilting toward 1-heavy tables and thereby breaking the output-negation symmetry. Two settings are explored below:

	β	α	μ
setting A (pure Occam)	1	0	0
setting B (colder, interface-averse, tilted)	1.5	0.5	−0.2

6.3 Trajectories

Figures 10–12 show the expected cumulative leverage $\langle L_\ell \rangle$ under both settings, for the uniform-order average and the strategy grid of Section 5.1. The ℓ-optimal pair targets the halfway horizon $\ell^* = Q/2$ and is plotted only up to it: past its own horizon the optimizer's continuation is not defined. Inside the optimal *set*, the fixed variant asks in random order. At $(4,1)$ only the contingent optimum is dropped – its state space is $(1+A)^Q = 3^{16} \approx 4 \cdot 10^7$ – while its fixed twin, a search over $\binom{16}{8} = 12{,}870$ sets, stays affordable. Two anchors hold in every panel: all schedules agree at $\ell = 1$, because input relabeling acts transitively on the questions and the Gibbs energies are relabeling-invariant, so single questions are exchangeable; and every full-length curve ends at the same $L_Q = H(M)_0/H(p)$, since any policy that asks everything receives exactly $H(p)$.

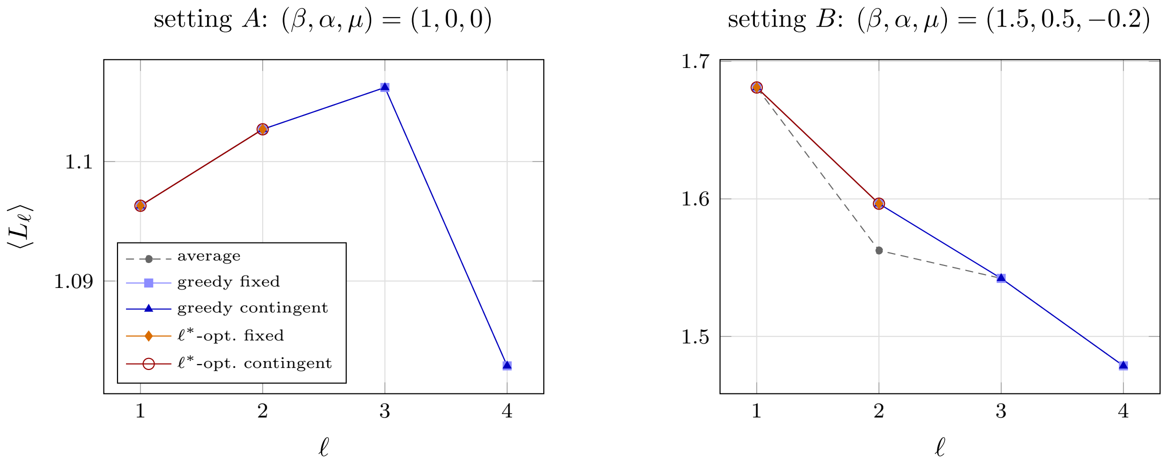


Figure 10: Expected leverage under the Gibbs prior at $(2, 1)$, $\ell^* = 2$. Left: under pure Occam all five schedules coincide exactly. Right: the fields split the schedules at $\ell = 2$.

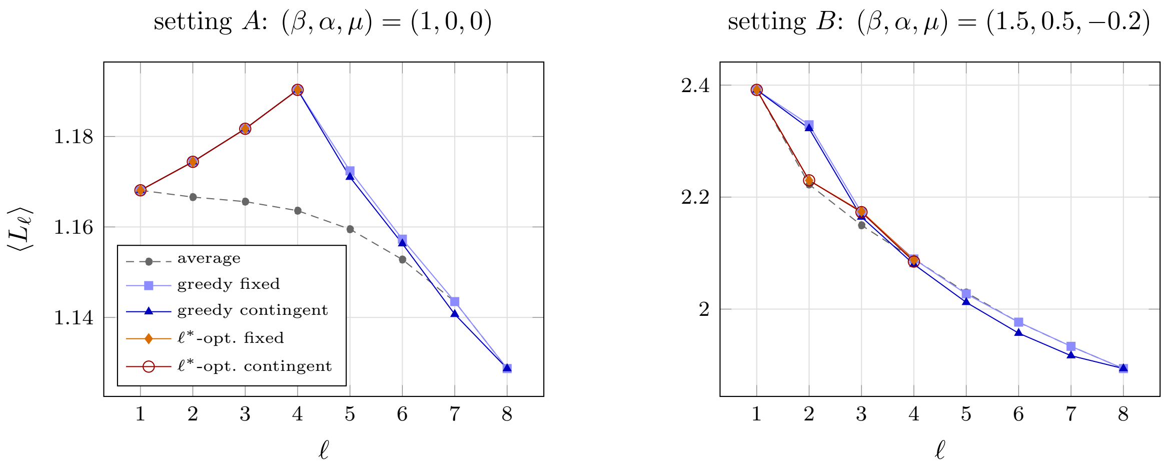


Figure 11: Expected leverage under the Gibbs prior at $(3, 2)$, $\ell^* = 4$. Right: at the horizon the ℓ^*-optimal schedules know the most and post the *lowest* leverage of the strategic four – knowledge and leverage are different objectives.

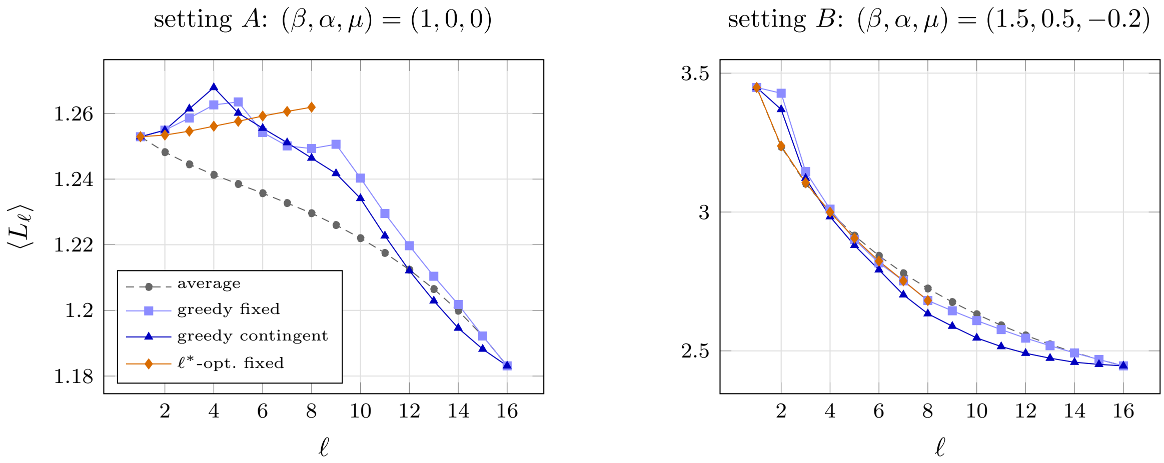


Figure 12: Expected leverage under the Gibbs prior at $(4, 1)$; only the contingent optimum is omitted. Under both settings the best fixed eight questions form a parity class of the input hypercube. The setting moves the curves by a factor of two and more; the schedule, by a few percent.

Three observations, one per system size.

At $(2, 1)$, under pure Occam, the strategy grid collapses *exactly*: all six question pairs share a single joint entropy (1.9679 bits), so every subset of every size is equivalent and every schedule is the uniform order. A complexity-only energy at this size creates correlation without preference. Setting B breaks the degeneracy – the footprint and bias fields split the pair orbits, and the strategic schedules lift above the average at $\ell = 2$ (1.5965 against 1.5625), greedy attaining the optimum. The scale jumps too: the colder ensemble stores $C_0 = 1.074$ bits against setting A's 0.306.

At $(3, 2)$, setting A rewards greed completely: the greedy fixed set is already the optimal set, and it is the four even-weight inputs $\{000, 011, 101, 110\}$ – a maximally spread design, the questions pairwise at Hamming distance two. Setting B stages the sharpest lesson of the section. The optimal set $\{001, 010, 100, 111\}$ differs from greedy's $\{000, 010, 100, 111\}$ by one question and knows 0.66 bits more at the horizon (expected remaining 1.26 against 1.92 bits) at an extra price of 0.32 received bits – and its leverage is *lower*: 2.088 against greedy's 2.090. The ℓ-optimal schedule maximizes knowledge, not leverage, and a ratio is perfectly happy with cheaper ignorance. Leverage measures how intelligently a learner spends, not how much they end up knowing.

At $(4, 1)$ the fixed optimum recovers the spread design at scale: under *both* settings the best eight questions are a parity class of the input hypercube – the even-weight inputs, pairwise Hamming distance two apart (input negation maps the class to its odd twin, an exactly equivalent design). Under setting B greedy-fixed lands on that odd twin, and the two curves merge at $\ell = 7, 8$; under pure Occam greedy's eight questions are *not* a parity class, and greedy both knows less at the horizon and posts the lower leverage (1.249 against the optimum's 1.262 – note the optimal curve is the only one that *rises* all the way to its horizon). Setting B pushes the ratio-versus-knowledge caution to its extreme: at $\ell = 8$ even the strategyless average out-levers the optimum (2.725 against 2.681) while knowing half as much (remaining 0.64 against 0.31 bits) – the typical random set is full of questions the cold prior half-knows already, so it pays 1.27 received bits where the maximally informative parity class pays 1.41. Beyond the schedules, the two settings differ far more than the schedules within either: setting A's curves sit within one percent of one another, while setting B – a cold ensemble in earnest, $H(p) = 1.68$ bits across 65,536 maps – runs from 3.45 down to

2.45. In the Gibbs world the dominant lever on leverage is the ensemble's coldness, not the question schedule.

7 The thermodynamic limit

The earlier chapters describe learning on a finite question set. This chapter asks what remains when the number of input bits tends to infinity. Since $Q = |\mathcal{Q}| = 2^n$, the limit $n \to \infty$ is one of an exponentially growing table, and we measure time by the fraction of that table already queried,

$$x \;=\; \frac{\ell}{Q} \;\in\; [0,1]. \tag{54}$$

The limit at fixed $x > 0$ is *macroscopic*: it records learning that occupies a nonzero fraction of the run. Learning completed in $O(1)$, in $O(\log Q)$, or more generally in $o(Q)$ questions is compressed into a boundary layer at $x = 0$. That distinction will explain both the hyperbolas below and several apparent noncommutations of limits.

Throughout, $u := A^{-1} = 2^{-m}$ abbreviates the reciprocal answer alphabet, and $N = A^Q$ counts the maps $\mathcal{Q} \to \mathcal{A}$, enumerated $\phi_0, \dots, \phi_{N-1}$ with prior weights $p_j = \Pr(\psi = \phi_j)$. One value of the truth ψ is drawn at the start of a run and supplies every answer in it; posteriors are conditional distributions of that same random map, and expected trajectories average over its prior law.

Fix an ordering $q_1, \dots, q_Q$ of the questions. For an index set $\mathcal{T} \subseteq \{1, \dots, Q\}$ write

$$\psi(\mathcal{T}) \;:=\; \big(\psi(q_k)\big)_{k\in\mathcal{T}} \;\in\; \mathcal{A}^{|\mathcal{T}|}, \qquad \Pr\big(\psi(\mathcal{T}) = y\big) \;=\; \sum_{j=0}^{N-1} p_j\,\mathbf{1}\big[\phi_j(\mathcal{T}) = y\big], \tag{55}$$

components in increasing order of k, with the parallel notation $\phi_j(\mathcal{T})$ for a fixed candidate map. The entropy $H(\psi(\mathcal{T}))$ measures uncertainty in those answers, not uncertainty about which questions were selected: $\mathcal{T}$ is held fixed. As in Chapter 5, the truth is drawn from the learner's prior and the questions are asked in uniformly random order, and leverage is always the ratio of expected totals, never the expectation of branchwise ratios. Binary entropy is written $h_2(z) = -z\log_2 z - (1-z)\log_2(1-z)$ with $0\log 0 := 0$; categorical entropies keep H.

7.1 A toy example: the spike prior and its renormalization flow

The spike prior places a distinguished weight on one map and spreads the remaining mass uniformly over every other map. Take the special map to be the all-zero map, $\phi_0(q) = 0^m$ for every $q \in \mathcal{Q}$, so that its answer to any k questions is the all-zero block 0^{mk}. This costs no generality: the answer alphabet may be relabelled separately at each question. Then

$$p_0 = p, \qquad p_j = \omega := \frac{1-p}{N-1} \quad (j \neq 0). \tag{56}$$

The plain $p = p_0$ is the mass of the all-zero map alone. This parameterization makes its uniform point $p = 1/N$, not $p = 0$; we normally consider $p \geq 1/N$, values below that describing an anti-spike.

The example is useful because it is closed under Bayesian conditioning. A confirming answer produces a sharper spike on a smaller map space; a refuting answer removes the special map and leaves equal weights on all survivors. The posterior therefore has a two-state flow, in which the uniform family is the stable, absorbing class: once a trajectory enters it, later conditioning keeps it there. An interior spike is a source of one-way probability current into that class, while conditional on avoiding refutation the surviving spike sharpens toward one.

7.1.1 The two-state flow and exact leverage

Let $\mathcal{T}_k := \{1, \ldots, k\}$ in the fixed ordering. A history remains on the spike branch for k steps exactly when $\psi(\mathcal{T}_k) = 0^{mk}$, and exactly $N_k := Nu^k$ maps produce that confirming history, one of which is ϕ_0. Its prior probability is therefore

$$P_k \;:=\; \Pr\left(\psi(\mathcal{T}_k) = 0^{mk}\right) \;=\; p + (N_k - 1)\,\omega \;=\; p + (Nu^k - 1)\,\omega. \tag{57}$$

On that event the posterior is another spike on those N_k maps,

$$p_k \;=\; \Pr\left(\psi = \phi_0 \mid \psi(\mathcal{T}_k) = 0^{mk}\right) \;=\; \frac{p}{P_k}, \qquad \omega_k \;=\; \frac{\omega}{P_k}, \tag{58}$$

so p_k is the time-indexed posterior height of the spike evaluated on the realized confirming branch, equal at $k = 0$ to the original weight p. For $0 \le k < Q$ the next zero answer is predicted by the special map and by $uN_k - 1$ of the surviving background maps, giving

$$\Pr\left(\psi(q_{k+1}) = 0^m \mid \psi(\mathcal{T}_k) = 0^{mk}\right) = p_k + (uN_k - 1)\,\omega_k = \frac{P_{k+1}}{P_k}, \qquad \boxed{P_k\,p_k \;=\; p} \tag{59}$$

The boxed product is the Bayesian consistency condition that organizes the example. The probability P_k of remaining on the spike branch decreases while the conditional height p_k increases, and their product stays equal to the original mass. Equivalently $\Pr(\psi = \phi_0 \mid \psi(\mathcal{T}_k))$ is a martingale under the Bayes average over histories: it equals p_k after k confirmations and zero after refutation, so its expectation is always $P_k p_k = p$. Renormalization sharpens the surviving spike without creating unconditional probability.

If any answer is nonzero then ϕ_0 is eliminated, every compatible survivor had the same mass ω, and the posterior is uniform; subsequent conditioning preserves uniformity. The two macrostates and their directions are therefore

$$\text{spike} \longrightarrow \begin{cases} \text{sharper spike,} & \text{with probability } P_{k+1}/P_k, \\ \text{uniform,} & \text{with probability } 1 - P_{k+1}/P_k, \end{cases} \qquad \text{uniform} \longrightarrow \text{uniform.} \tag{60}$$

For $0 \le k < Q$ and $p < 1$ the exact odds on the surviving branch are $p_k/(1 - p_k) = p(N-1)/\big[(1-p)(Nu^k - 1)\big]$, and for fixed $p > 0$,

$$P_k = p + (1-p)u^k + O(N^{-1}), \qquad p_k = \frac{p}{p + (1-p)u^k} + O(N^{-1}). \tag{61}$$

If in addition $0 < p < 1$ and $u^{\,Q-k} \to 0$, the finite-population correction disappears and $p_k/(1 - p_k) \simeq u^{-k}p/(1-p)$: each confirmation supplies approximately $-\log_2 u = m$ bits of evidence for the spike. Because $N = u^{-Q}$, equation (57) gives $P_Q = p$ and hence $p_Q = 1$; the histories that confirm through the complete table are precisely those with $\psi = \phi_0$. At the uniform point $p = 1/N$ both posterior branches are uniform on their surviving map spaces, and at $p = 1$ the delta state is absorbing. For $1/N < p < 1$, false-spike histories flow irreversibly into the uniform class while the confirming history sharpens toward the delta.

The same P_k has a second reading: it is the probability of the single k-answer block 0^{mk}, so there is no new distribution to compute. For $1 \le k \le Q$ every other answer block is realized by N_k background maps and therefore carries the common probability

$$U_k \;:=\; \frac{1 - P_k}{u^{-k} - 1} \;=\; N_k\,\omega, \qquad \text{so} \qquad \Pr(\psi(\mathcal{T}) = y) = \begin{cases} P_k, & y = 0^{mk}, \\ U_k, & y \neq 0^{mk}, \end{cases} \tag{62}$$

for any index set $\mathcal{T}$ of size k. The entropy of that block distribution is

$$G_k \;=\; -P_k \log_2 P_k - (u^{-k} - 1)\,U_k \log_2 U_k. \tag{63}$$

Every choice of k question indices obeys the same law, so (63) is also the mean k-question block entropy of Chapter 5. In particular $G_0 = 0$, $G_Q = H(\psi)$ and $H(M)_0 = Q\,G_1$, with the exact prior entropy

$$G_Q \;=\; h_2(p) + (1-p)\log_2(N-1), \tag{64}$$

extensive in Q at fixed $0 < p < 1$ and vanishing at $p = 1$.

The two-state flow now gives the finite trajectory without another posterior calculation. The entropy chain rule says that joint uncertainty is the entropy of what is observed first plus the average uncertainty left in what is observed next, so $G_{k+1} - G_k$ is the expected conditional entropy of one fresh answer after k answers have been seen. Hence the expected information received is G_k, each of the $Q - k$ unasked questions carries mean conditional entropy $G_{k+1} - G_k$, and

$$\mathbb{E}[H(M_k)] = (Q-k)(G_{k+1} - G_k), \qquad \boxed{\langle L_k \rangle = \frac{Q\,G_1 - (Q-k)(G_{k+1} - G_k)}{G_k}} \tag{65}$$

for $1 \le k < Q$ and $p < 1$, with $\langle L_Q \rangle = QG_1/G_Q$ at completion. One counting formula, (57), thus determines the Bayesian flow, the answer-block entropy, and the exact leverage; no separate finite-trajectory ansatz is required.

7.1.2 One question, branch by branch

For a single question abbreviate $P := P_1$ and $U := U_1$, so that $P + (A-1)U = 1$: the zero answer has probability P, while each particular nonzero answer has probability U. Here *correct* means that the observed answer agrees with the spike map at this question; it does not assert that the entire spike map is the truth. The surprisals are

$$s_{\text{correct}} = -\log_2 P, \qquad s_{\text{wrong}} = -\log_2 U, \qquad \mathbb{E}[s] = P\,s_{\text{correct}} + (A-1)U\,s_{\text{wrong}} = G_1, \tag{66}$$

a particular wrong answer having probability U, not $1 - P$. Before asking, every one of the Q columns carries the same answer distribution $(P, U, \ldots, U)$ and hence entropy G_1, so $H_{\text{before}} = H(M)_0 = Q\,G_1$. Conditional on confirmation the next unasked answer has probabilities P_2/P for zero and U_2/P for each nonzero symbol, with entropy

$$H_{\text{c}} \;:=\; -\frac{P_2}{P}\log_2\frac{P_2}{P} - (A-1)\frac{U_2}{P}\log_2\frac{U_2}{P}, \qquad H_{\text{after,correct}} = (Q-1)H_{\text{c}}, \tag{67}$$

the asked column being fixed while all $Q-1$ unasked columns share that conditional marginal. Refutation instead leaves equal posterior weights on all compatible maps, so every unasked answer is uniform on A symbols and $H_{\text{after,wrong}} = (Q-1)m$. Weighting the two branches,

$$\mathbb{E}[H(M_1)] = P\,H_{\text{after,correct}} + (A-1)U\,H_{\text{after,wrong}} = (Q-1)(G_2 - G_1), \tag{68}$$

the first equality ordinary branch averaging, the second the $k = 1$ case of (65).

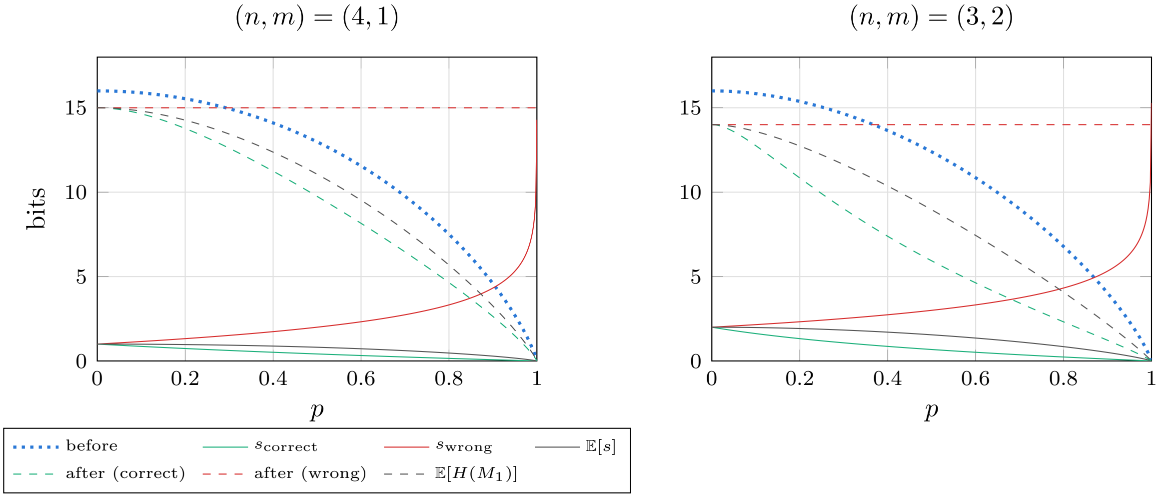


Figure 13: One-question anatomy of the spike prior against the special map's mass p: the branch surprisals (66) (solid), the answer-matrix entropy before the question (dotted) and on each branch after it (67) (dashed). Both panels carry $Qm = 16$ bits of table entropy at the uniform point.

For $p < 1$ the branch leverages divide the vertical distance between the pre-question curve and each branch curve by that branch's own surprisal,

$$L_{\text{correct}} = \frac{QG_1 - (Q-1)H_c}{-\log_2 P}, \qquad L_{\text{wrong}} = \frac{QG_1 - (Q-1)m}{-\log_2 U}, \tag{69}$$

whereas the expected leverage is the ratio of the two expected totals,

$$\boxed{\langle L_1 \rangle = \frac{P\big[QG_1 - (Q-1)H_c\big] + (A-1)U\big[QG_1 - (Q-1)m\big]}{G_1} = \frac{QG_1 - (Q-1)(G_2 - G_1)}{G_1}.} \tag{70}$$

The branch formulas use realized surprisals; the expected formula divides the expected entropy drop by the expected surprisal G_1 and is therefore *not* the probability-weighted average of the two branch ratios.

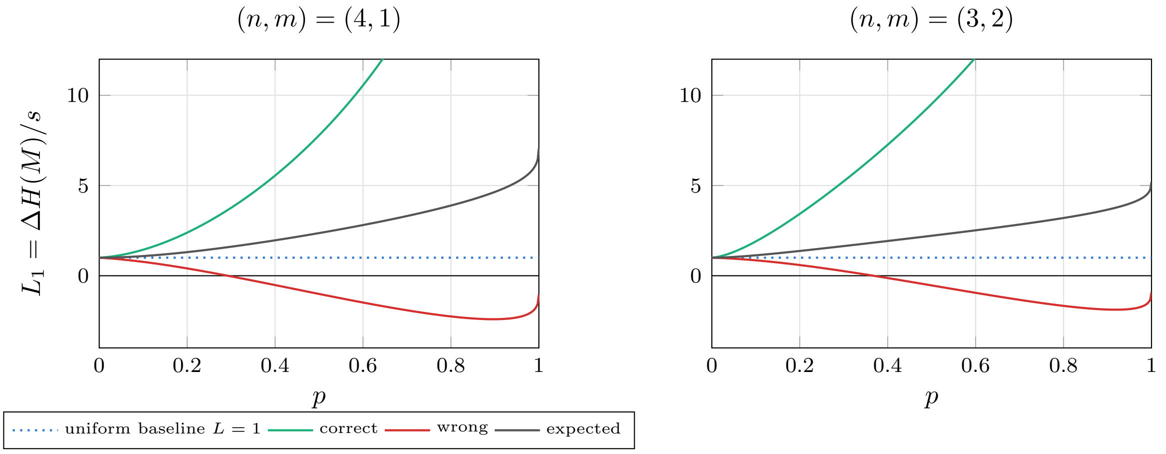


Figure 14: Branchwise (69) and expected (70) leverage of the first question. A refuting answer carries so much surprisal that its own ratio falls below the uniform baseline and eventually below zero, yet the expectation stays well above 1 because the confirming branch is both likely and cheap.

7.1.3 The extraction ratio

The total correlation $C := H(M)_0 - G_Q = QG_1 - G_Q$ is the information stored in dependencies among answers rather than visible in their separate marginals. Only the part of the expected entropy drop beyond the received entropy G_1 comes out of that store, so when $C > 0$ the share of it freed by a single question is

$$\boxed{R_1 \;:=\; \frac{\mathbb{E}[\Delta H(M)] - G_1}{C} \;=\; \frac{(Q-1)(2G_1 - G_2)}{QG_1 - G_Q}.} \tag{71}$$

The two ratios of this chapter differ only in their denominator: R_1 measures against the whole dependency stock, leverage against the received surprisal, and

$$R_1 = \frac{G_1}{C}\big(\langle L_1\rangle - 1\big), \qquad \langle L_1\rangle = 1 + \frac{C}{G_1}R_1. \tag{72}$$

Thus R_1 is a stock-extraction fraction while $\langle L_1\rangle - 1$ is released dependency information per received bit. At $p = 1$ both the numerator of R_1 and C vanish; expanding (63) with $\varepsilon = 1 - p \to 0^+$ gives the exact finite-(n, m) ratio limit

$$\boxed{\lim_{p\to 1^-} R_1(p) = \frac{(Q-1)N(1-u)^2}{QN(1-u) - (N-1)},} \qquad \frac{(Q-1)(1-u)^2}{Q(1-u) - 1} \tag{73}$$

the second expression being the additional large-N approximation. At $(n, m) = (2, 1)$ the exact value is $12/17 \simeq 0.706$ against the approximation's 0.75. Both share the thermodynamic limit

$$\boxed{\lim_{n\to\infty}\lim_{p\to 1^-} R_1(p) \;=\; 1 - u \;=\; 1 - 2^{-m}.} \tag{74}$$

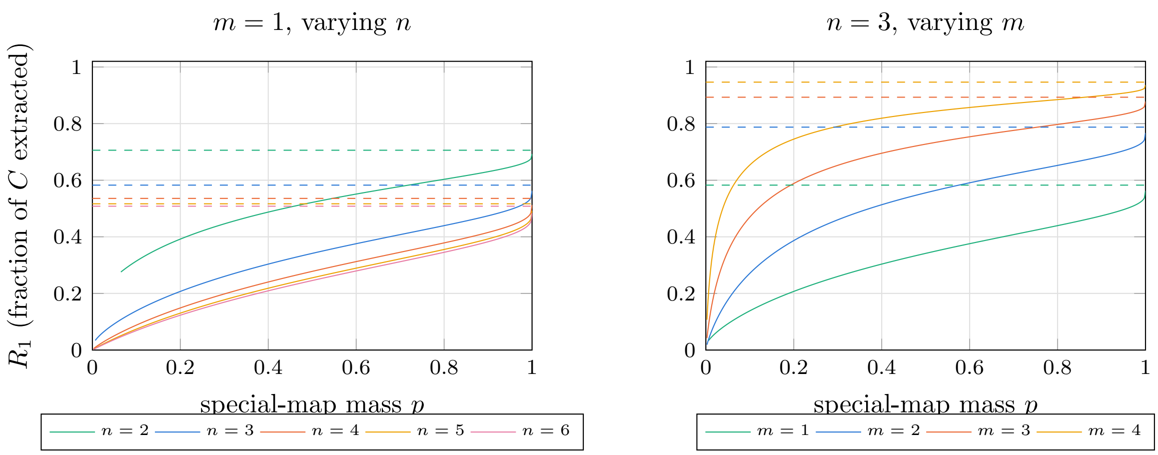


Figure 15: The exact one-question extraction ratio (71) of the spike prior, plotted only where it is defined ($1/N < p < 1$). Dashed lines are the exact finite-N limits (73) as $p \to 1^-$. Growing n drives them down to $1 - u = 1/2$; growing m lifts them toward one.

For the spike-plus-uniform family (74) is a theorem. Extending it to a broader class of uninformative hedges would require a precise definition and a separate extremal argument: full support alone is not sufficient.

7.1.4 The sharp-prior and thermodynamic limits

The sharp-prior calculation takes $\varepsilon = 1 - p \to 0^+$ with n and m fixed. It is a limit toward the delta prior, not evaluation at $p = 1$, where leverage is $0/0$. With the entropy coefficient $\iota_k := \frac{N}{N-1}(1 - u^k)$ we have $P_k = 1 - \varepsilon\iota_k$, the other $u^{-k} - 1$ outcomes share mass $\varepsilon\iota_k$, and

$$G_k = h_2(\varepsilon\iota_k) + \varepsilon\iota_k \log_2(u^{-k} - 1) \;\sim\; \varepsilon\,\iota_k \log_2 \frac{1}{\varepsilon} \qquad (\varepsilon \downarrow 0) \tag{75}$$

at fixed finite Q and k. Substituting into (65) cancels the common factor $\varepsilon \log_2(1/\varepsilon)$ and leaves, for $1 \le k \le Q$,

$$\boxed{\lim_{p\to 1^-} \langle L_k\rangle = (1-u)\,\frac{Q - (Q-k)u^k}{1 - u^k} = \underbrace{Q(1-u)}_{\text{extensive plateau}} + \underbrace{(1-u)\frac{k\,u^k}{1-u^k}}_{O(1)\text{ boundary layer}}\,.} \tag{76}$$

The second term decays geometrically in k and does not depend on Q at all. In particular $\lim_{p\to 1^-}\langle L_1\rangle = Q - (Q-1)2^{-m}$, so

$$\boxed{\lim_{n\to\infty}\lim_{p\to 1^-} \frac{\langle L_1\rangle}{Q} = 1 - 2^{-m}.} \tag{77}$$

The coefficient $1 - 2^{-m}$ is the fraction of the hypothesis count eliminated by one answer. Its equality with the normalized leverage follows from the entropy asymptotics and should not be read as literally settling that fraction of every individual column. The limiting operation is necessary: at $p = 1$ the learner already assigns probability one to the truth, so received and destroyed information both vanish and leverage is $0/0$. The finite value is the ratio at which the two vanish as $p \to 1^-$. Taking $k = \lfloor xQ \rfloor$ in (76) gives the same constant at every $x > 0$, with the uniform bound

$$\sup_{1\le k\le Q} \left| \frac{1}{Q} \lim_{p\to 1^-} \langle L_k\rangle - (1-u) \right| = O(Q^{-1}), \tag{78}$$

so the microscopic boundary term survives only as an $O(1)$ correction to the unnormalized leverage.

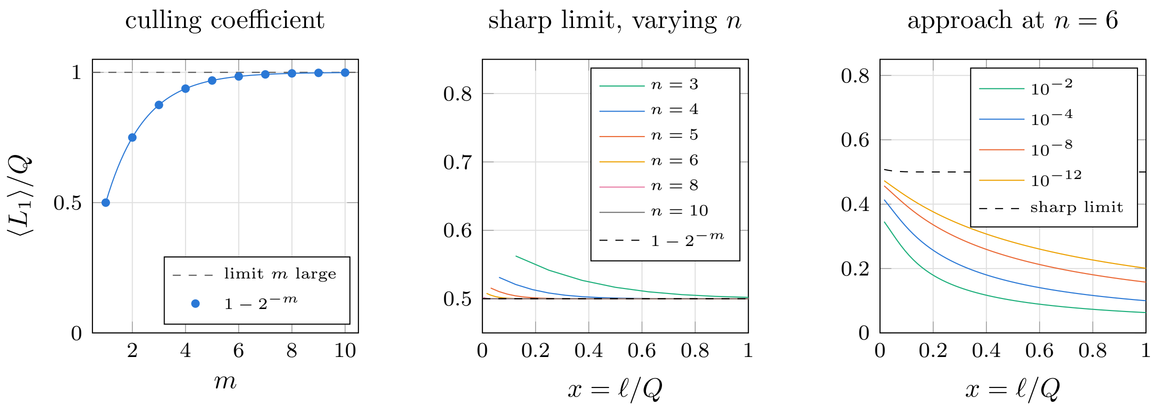


Figure 16: Left: the culling coefficient (77) against the answer width. Middle: the sharp-prior curve (76) normalized by Q, whose $O(1)$ boundary layer is squeezed into the origin as n grows, leaving the plateau at $1-u$. Right: the approach at fixed $n = 6$, $m = 1$, labelled by $1 - p$. Tightening the prior lifts the curve only logarithmically, so the plateau is approached pointwise in x but never uniformly.

The other order of limits has a different scale. Fix $0 < p < 1$ and $x > 0$, then send $n \to \infty$. The limiting probability of the spike answer in one column is $p + (1-p)u$, so the one-column entropy is

$$h_{\text{spike}}(p, m) := h_2\big(p + (1-p)u\big) + (1-p)(1-u)\log_2(A-1). \tag{79}$$

At any fixed $x > 0$ the conditional height p_k on a surviving branch has approached one within a vanishing fraction of the run. The branch itself still occurs with limiting probability p, and the complementary probability $1-p$ lies on uniform refuted branches, so $\mathbb{E}[H(M_{\lfloor xQ \rfloor})]/Q \to (1-x)(1-p)m$ and $G_{\lfloor xQ \rfloor}/Q \to x(1-p)m$. Since $G_1 \to h_{\text{spike}}(p, m)$, dividing the numerator and denominator of (65) by Q gives

$$\boxed{L_p(x) = \frac{h_{\text{spike}}(p,m) - (1-x)(1-p)m}{x(1-p)m} = 1 + \frac{h_{\text{spike}}(p,m) - (1-p)m}{x(1-p)m}, \qquad 0 < x \le 1.} \tag{80}$$

The $1/x$ term records information released in a vanishing initial fraction of the run, divided by the $O(xQ)$ information received afterwards.

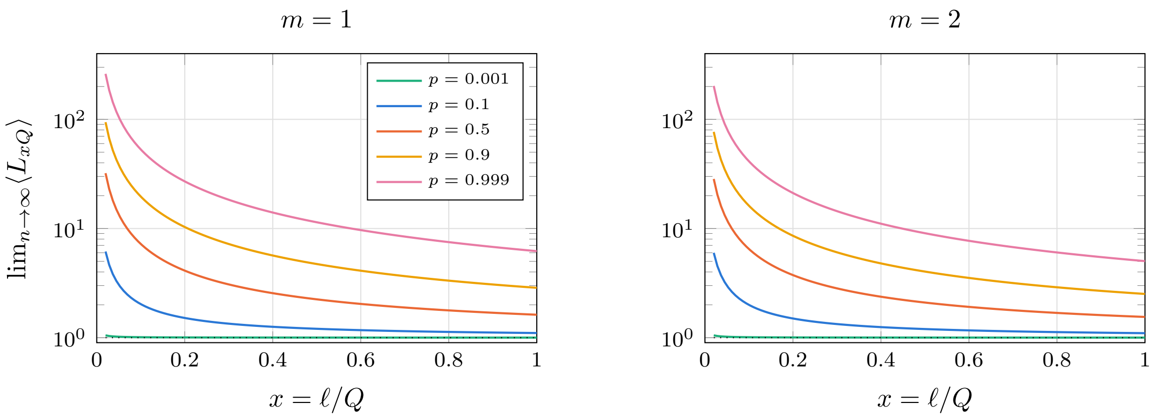


Figure 17: The fixed-p thermodynamic curve (80), on a logarithmic vertical scale. Every weight produces the same $1/x$ shape; the prior sets only the coefficient, which grows from 0.001 at $p = 0.001$ to above 5 at $p = 0.999$.

As $p \to 1^-$ that coefficient diverges, $\big[h_{\text{spike}}(p,m) - (1-p)m\big]/\big[(1-p)m\big] \sim \frac{1-2^{-m}}{m}\log_2\frac{1}{1-p}$, which is the noncommutation in explicit form. Writing $L_{p,n}(x) := \langle L_{\lfloor xQ \rfloor}\rangle$ and comparing under the same normalization,

$$\lim_{n\to\infty}\lim_{p\to1^-}\frac{L_{p,n}(x)}{Q} = 1 - 2^{-m}, \qquad \lim_{p\to1^-}\lim_{n\to\infty}\frac{L_{p,n}(x)}{Q} = 0. \tag{81}$$

Before dividing by Q the first ordering is extensive while the second diverges only logarithmically in $1/(1-p)$. The two orders are the endpoints of a family of joint scalings. Choose $p_n \to 1$ with prescribed exponential sharpness

$$\frac{\log_2\big(1/(1-p_n)\big)}{Q} \longrightarrow \lambda \in (0, \infty), \tag{82}$$

so that $\lambda = 0$ recovers the scaling behind the second limit and the formal endpoint $\lambda = \infty$ the first. Putting $\varepsilon_n = 1 - p_n$, the leading entropies for $0 < x < 1$ and $k = \lfloor xQ \rfloor$ are

$G_1 \sim (1-u)\varepsilon_n \lambda Q$, $G_k \sim \varepsilon_n Q(\lambda + mx)$ and $G_{k+1} - G_k \sim \varepsilon_n m$, and substitution into (65) gives the crossover

$$\boxed{\frac{L_{p_n,n}(x)}{Q} \longrightarrow (1-u)\,\frac{\lambda}{\lambda + mx}.} \tag{83}$$

At $x = 1$ the increment $G_{k+1} - G_k$ is undefined and the same value follows instead from the completion formula. The two familiar orders are thus endpoints of a larger family.

7.2 Thermodynamic-limit calculus

The spike is solvable because all subsets of the same size have the same entropy. The general theory replaces that symmetry by an average over subsets.

7.2.1 The finite sequence from which the limit is taken

For each fixed $\mathcal{T} \subseteq \{1, \ldots, Q\}$ the vector $\psi(\mathcal{T})$ collects the answers at those indices, and its entropy is taken under the prior law of ψ with $\mathcal{T}$ held fixed. Define

$$G_{n,k} \;:=\; \binom{Q}{k}^{-1} \sum_{\substack{\mathcal{T}\subseteq\{1,\ldots,Q\}\\ |\mathcal{T}|=k}} H\big(\psi(\mathcal{T})\big), \qquad 0 \le k \le Q, \tag{84}$$

so that $G_{n,0} = 0$ and $G_{n,Q} = H(\psi) = H(p^{(n)})$, where $p^{(n)}$ is the prior law of the random map at size n, with increments $\gamma_{n,k} := G_{n,k+1} - G_{n,k}$.

The entropy chain rule separates uncertainty already present in one variable from what is left in another once the first is known: for discrete X_1, X_2,

$$H(X_1, X_2) = H(X_1) + H(X_2 \mid X_1), \qquad H(X_2 \mid X_1) = \sum_x \Pr(X_1 = x)\, H(X_2 \mid X_1 = x). \tag{85}$$

Applying it to $\psi(\mathcal{T})$ and one fresh answer $\psi(q_i)$ with $i \notin \mathcal{T}$, and averaging over the externally selected indices, gives

$$\gamma_{n,k} \;=\; \mathbb{E}_{\mathcal{T},i}\big[H\big(\psi(q_i) \mid \psi(\mathcal{T})\big)\big]. \tag{86}$$

There are two separate averages here: the conditional entropy averages over answer histories under the prior on the random map, while $\mathbb{E}_{\mathcal{T},i}$ averages that entropy over which questions were selected. Conditioning reduces entropy, so coupling a random k-set to a random $(k{+}1)$-set yields

$$m \;\ge\; \gamma_{n,0} \;\ge\; \gamma_{n,1} \;\ge\; \cdots \;\ge\; \gamma_{n,Q-1} \;\ge\; 0. \tag{87}$$

This monotonicity is deterministic, the truth and the subset having already been averaged over. It should not be called self-averaging; concentration of individual learning trajectories would be a separate result. The exact finite-n identities are then

$$\mathbb{E}[\text{received through } \ell] = G_{n,\ell}, \qquad \mathbb{E}[\text{remaining table entropy}] = (Q-\ell)\,\gamma_{n,\ell}, \tag{88}$$

for $0 \le \ell < Q$, with zero remaining entropy at $\ell = Q$, so

$$\boxed{\langle L_{n,\ell}\rangle = \frac{Q\,G_{n,1} - (Q-\ell)\,\gamma_{n,\ell}}{G_{n,\ell}}} \qquad (1 \le \ell < Q), \qquad \langle L_{n,Q}\rangle = \frac{Q\,G_{n,1}}{G_{n,Q}}. \tag{89}$$

7.2.2 A precise macroscopic-limit statement

It is useful to keep the initial marginal entropy separate from the macroscopic profile on the right, since that allows an $o(Q)$-question boundary layer at $x = 0$. Let $h_0 := \lim_{n\to\infty} G_{n,1}$, and suppose the staircases have an almost-everywhere limit γ on $(0, 1)$, with $\gamma_{n,\lfloor xQ\rfloor} \to \gamma(x)$ at any particular x where the remaining entropy is evaluated. At such a point $\gamma(x)$ is the expected entropy of one more answer after a random x-fraction has been observed. Four hypotheses suffice:

1. **Fixed answer width.** The integer m is fixed, which supplies the uniform bound $0 \le \gamma_{n,k} \le m$.
2. **Initial-entropy convergence.** The limit h_0 exists.
3. **Profile convergence.** The staircase $t \mapsto \gamma_{n,\lfloor tQ\rfloor}$ converges almost everywhere on $(0, 1)$, with direct convergence at any x where the remaining entropy is evaluated. At continuity points of a selected monotone limit this causes no ambiguity.
4. **Positive entropy density.** The limiting area is positive, $\int_0^1 \gamma(t)\, dt > 0$.

The fourth condition is the nondegenerate form of extensivity, and it implies $G_{n,Q}/Q \to \int_0^1 \gamma > 0$. Merely writing $G_{n,Q} = \Theta(Q)$ is not sufficient: the entropy density may oscillate with n, and even a convergent scalar density does not determine the profile. Conversely, profile convergence with positive area implies the extensivity required. No Gibbs representation and no full-support lower bound on individual map probabilities is needed.

Extend the finite increments to a staircase on $[0, 1]$ by $\gamma_n(t) := \gamma_{n,\min(\lfloor tQ\rfloor, Q-1)}$, the value at the single endpoint $t = 1$ being irrelevant to the integral. Telescoping gives an exact Riemann-sum identity at grid points,

$$\frac{G_{n,\ell}}{Q} = \frac{1}{Q}\sum_{k=0}^{\ell-1} \gamma_{n,k} = \int_0^{\ell/Q} \gamma_n(t)\, dt, \tag{90}$$

and since $0 \le \gamma_n \le m$, dominated convergence gives $G_{n,\ell_n}/Q \to g(x) := \int_0^x \gamma$ for $\ell_n = \lfloor xQ\rfloor$, while at a point of profile convergence $(Q - \ell_n)\gamma_{n,\ell_n}/Q \to (1 - x)\gamma(x)$. Substituting both into (89) proves

$$\boxed{L(x) \;=\; \frac{h_0 - (1 - x)\,\gamma(x)}{g(x)}, \qquad g(x) = \int_0^x \gamma(t)\, dt, \qquad 0 < x < 1.} \tag{91}$$

Since γ is nonnegative and nonincreasing, a positive integral makes $g(x) > 0$ for every $x > 0$.

Monotonicity and boundedness guarantee convergent subsequences by a Helly selection argument, but they do not guarantee that the full sequence of priors selects a unique profile: a family can alternate between two constructions on even and odd n. At the prior level one therefore needs a coherent construction, such as a fixed exchangeable directing measure or fixed local block types with convergent frequencies, or must take profile convergence itself as a hypothesis. Endpoint expansions, boundary-layer singularities and discontinuous profiles are collected in Appendix H.

7.3 The coin-mixture prior

The simplest exchangeable scaling family is a mixture of independent-answer models. It gives a complete calculus exercise and a clean example of the boundary layer at $x = 0$. Throughout this section the answer width m and the component family are fixed while $n \to \infty$.

7.3.1 Categorical formulation

Let Z take values in a finite set of components indexed by r, with $\Pr(Z = r) = w_r$ summing to one. The mixture defines a joint distribution for the component Z and the random map ψ, the component being sampled once for the entire map rather than once per question. Conditional on $Z = r$ the answers at distinct question indices are independent with categorical distribution ν_r on the A possible answers:

$$\nu_r : \mathcal{A} \to [0,1], \qquad \sum_{a\in\mathcal{A}} \nu_r(a) = 1, \qquad \Pr\big(\psi(q_i) = a \mid Z = r\big) = \nu_r(a). \tag{92}$$

Thus ν_r acts on *one possible answer* a, not on a whole map: it returns the probability that component r assigns to that answer at any question. For $m = 1$ a Bernoulli component with bias θ_r has $\nu_r(1) = \theta_r$; for $m = 2$ a component could instead be the table $\nu_r(00), \dots, \nu_r(11) = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$, a distribution on the four two-bit answers that need not factor into independent bits. The joint law is

$$\Pr(Z = r, \psi = \phi_j) = w_r \prod_{i=1}^{Q} \nu_r\big(\phi_j(q_i)\big), \qquad p_j = \sum_r w_r \prod_{i=1}^{Q} \nu_r\big(\phi_j(q_i)\big), \tag{93}$$

the product being the map's probability conditional on component r and the outer sum averaging those likelihoods. Write $\nu_{\mathrm{mix}} := \sum_r w_r \nu_r$ for the one-question mixture, so that

$$h_0 = H(\nu_{\mathrm{mix}}), \qquad h_{\mathrm{cond}} := \sum_r w_r H(\nu_r) \tag{94}$$

are the entropy of one answer before anything is known and the entropy per question once the component is known.

Because the construction treats all questions alike, $H(\psi(\mathcal{T}))$ takes the same value G_k for every $\mathcal{T}$ of size k. Applying the chain rule to the joint distribution of Z and $\psi(\mathcal{T})$ in both orders,

$$H\big(Z, \psi(\mathcal{T})\big) = H(Z) + H\big(\psi(\mathcal{T}) \mid Z\big) = H\big(\psi(\mathcal{T})\big) + H\big(Z \mid \psi(\mathcal{T})\big), \tag{95}$$

and using conditional independence, $H(\psi(\mathcal{T}) \mid Z) = k\, h_{\mathrm{cond}}$, the two expansions rearrange into the exact decomposition

$$G_k = k\, h_{\mathrm{cond}} + I_k, \qquad I_k := I\big(Z; \psi(\mathcal{T})\big), \qquad \gamma_k = h_{\mathrm{cond}} + (I_{k+1} - I_k), \tag{96}$$

in which I_k is the mutual information between the selected component and the first k answers. The correction has the conditional-information form $I_{k+1} - I_k = I(Z; \psi(q_i) \mid \psi(\mathcal{T})) \geq 0$, exactly the additional information that the next answer reveals about which component was selected. Substituting into (89) gives the exact finite-n leverage

$$\langle L_{n,\ell} \rangle = \frac{Q\, H(\nu_{\mathrm{mix}}) - (Q - \ell)\big[h_{\mathrm{cond}} + I_{\ell+1} - I_\ell\big]}{\ell\, h_{\mathrm{cond}} + I_\ell}, \tag{97}$$

and, provided $h_{\mathrm{cond}} > 0$, the inequalities $I_\ell \geq 0$ and $I_{\ell+1} - I_\ell \geq 0$ show that the finite curve lies below its limiting hyperbola at the same $x = \ell/Q$.

For finitely many components $0 \leq I_k \leq H(Z) < \infty$: the information required to identify the component is subextensive. Discard any zero-weight components; if the remaining distinct components are identifiable and mutually absolutely continuous, then for data generated from a positive-weight component r and any positive-weight competitor s, almost surely under $\nu_r^{\otimes\infty}$,

$$\frac{1}{k} \log_2 \frac{\Pr(Z = s \mid \psi(\mathcal{T}_k))}{\Pr(Z = r \mid \psi(\mathcal{T}_k))} \;\longrightarrow\; -D_{\mathrm{KL}}(\nu_r \| \nu_s) \;<\; 0, \tag{98}$$

with D_{KL} in base-two logarithms. With a support mismatch the divergence may be $+\infty$, and one revealing symbol can instead set the wrong component's posterior probability to zero. Either way the posterior predictive converges to the true ν_r, and bounded convergence gives $\gamma(x) = h_{\mathrm{cond}}$ for every fixed $0 < x < 1$. Then $g(x) = x\, h_{\mathrm{cond}}$ and (91) collapses to

$$\boxed{L(x) = 1 + \frac{c}{x}, \qquad c = \frac{H(\nu_{\mathrm{mix}}) - h_{\mathrm{cond}}}{h_{\mathrm{cond}}} \ \geq\ 0,} \tag{99}$$

the inequality being the concavity of entropy, $H(\sum_r w_r \nu_r) \geq \sum_r w_r H(\nu_r)$. Equality holds exactly when all positive-weight components share the same categorical distribution once duplicates are merged, the effective one-coin model with $L \equiv 1$. If instead $h_{\mathrm{cond}} = 0$, as for a nontrivial mixture of deterministic coins, the map entropy is $O(1)$ and bounded by $H(Z)$ rather than extensive in Q; the normalization of Section 7.2 degenerates and (99) is not a finite thermodynamic curve.

The similarity to the fixed-p spike is now explicit. In both cases a global latent variable is inferred within $o(Q)$ questions, leaving a constant positive entropy density afterwards. The macroscopic profile sees only the jump from h_0 down to h_{cond}, and equation (165) of Appendix H turns that jump into $1/x$.

7.3.2 The binary coin and exact finite formulas

For $m = 1$ write $\nu_r = \mathrm{Bernoulli}(\theta_r)$ and denote the truth-table Hamming weight by $|\phi_j| := \sum_i \phi_j(q_i)$, so that $p_j = \sum_r w_r \theta_r^{|\phi_j|}(1-\theta_r)^{Q-|\phi_j|}$. After ℓ observations containing k ones, let

$$P_{\ell,k} := \sum_r w_r\, \theta_r^k (1-\theta_r)^{\ell-k}. \tag{100}$$

Every particular binary string of length ℓ and weight k has this probability, so

$$G_\ell = -\sum_{k=0}^{\ell} \binom{\ell}{k} P_{\ell,k} \log_2 P_{\ell,k}, \qquad w_r(\ell,k) = \frac{w_r \theta_r^k (1-\theta_r)^{\ell-k}}{P_{\ell,k}}, \tag{101}$$

the posterior component weights being defined for count classes with $P_{\ell,k} > 0$. Consequently the next-answer probability is $\Pr(\psi(q_i) = 1 \mid \psi(\mathcal{T}) = y) = \sum_r w_r(\ell,k)\theta_r$ for $|\mathcal{T}| = \ell$, $i \notin \mathcal{T}$ and y of weight k, and averaging its binary entropy over all count classes gives

$$\gamma_\ell = \sum_{\substack{0 \leq k \leq \ell \\ P_{\ell,k} > 0}} \binom{\ell}{k} P_{\ell,k}\, h_2\!\left(\sum_r w_r(\ell,k)\theta_r\right) = G_{\ell+1} - G_\ell. \tag{102}$$

Zero-probability terms in (101) use $0 \log 0 = 0$, while (102) simply omits those classes, whose posterior weights are undefined and whose contribution vanishes. Equation (101) is valid through $\ell = Q$ and (102) through $\ell = Q - 1$; apart from that truncation, neither depends on n.

EXAMPLE

Take $(\theta_1, \theta_2) = (0.1, 0.9)$ with equal weights, so $Q = 4$ at $(n, m) = (2, 1)$. For a length-ℓ string of weight k, equation (100) becomes $P_{\ell,k} = (9^k + 9^{\ell-k})/(2 \cdot 10^\ell)$, so a map's probability depends only on its Hamming weight, with clearly separated masses:

weight $\lvert\phi_j\rvert$	0	1	2
mass of each map	0.3281	0.0369	0.0081
multiplicity	1	4	6

with weights 3 and 4 mirroring weights 1 and 0. A weight-one map is thus more than four times as probable as a weight-two map, and the normalization $2(0.3281) + 8(0.0369) + 6(0.0081) = 1$ is visible directly. In the truth-table ordering of the earlier examples:

j	p_j	j	p_j	j	p_j	j	p_j
0000	0.3281	0001	0.0369	0010	0.0369	0011	0.0081
0100	0.0369	0101	0.0081	0110	0.0081	0111	0.0369
1000	0.0369	1001	0.0081	1010	0.0081	1011	0.0369
1100	0.0081	1101	0.0369	1110	0.0369	1111	0.3281

The mixture mean is $\frac{1}{2}(0.1) + \frac{1}{2}(0.9) = \frac{1}{2}$, so $G_1 = h_2(1/2) = 1$. If the first answer is 1, Bayes' rule moves the component weights from $(\frac{1}{2}, \frac{1}{2})$ to $(0.1, 0.9)$ and the next-one probability becomes $0.1(0.1) + 0.9(0.9) = 0.82$; if it is 0 the weights reverse and that probability is 0.18. Both branches therefore have predictive entropy $h_2(0.18) = h_2(0.82)$, so $G_2 - G_1 = h_2(0.18) = 0.680077$. The remaining block entropies follow from the same count-by-weight formula: for two answers the strings $00, 11$ each have probability 0.41 and $01, 10$ each 0.09; for three answers the two constant strings each have 0.365 and the other six each 0.045. Hence

$$(G_0, \ldots, G_4) = (0,\ 1,\ 1.680077,\ 2.269405,\ 2.797921), \tag{103}$$

with increments $(1,\ 0.680077,\ 0.589327,\ 0.528517)$ and exact cumulative leverages 1.95977, 1.67930, 1.52969, 1.42963. The conditional entropy density and hyperbola coefficient are $h_{\text{cond}} = h_2(0.1) = 0.468996$ and $c = (1 - h_2(0.1))/h_2(0.1) = 1.132216$, so the limiting curve is $L(x) = 1 + 1.132216/x$. The small-n curve lies well below it because four observations cannot identify the latent coin sharply: its endpoint is 1.42963 against the limiting $1 + c = 2.13222$.

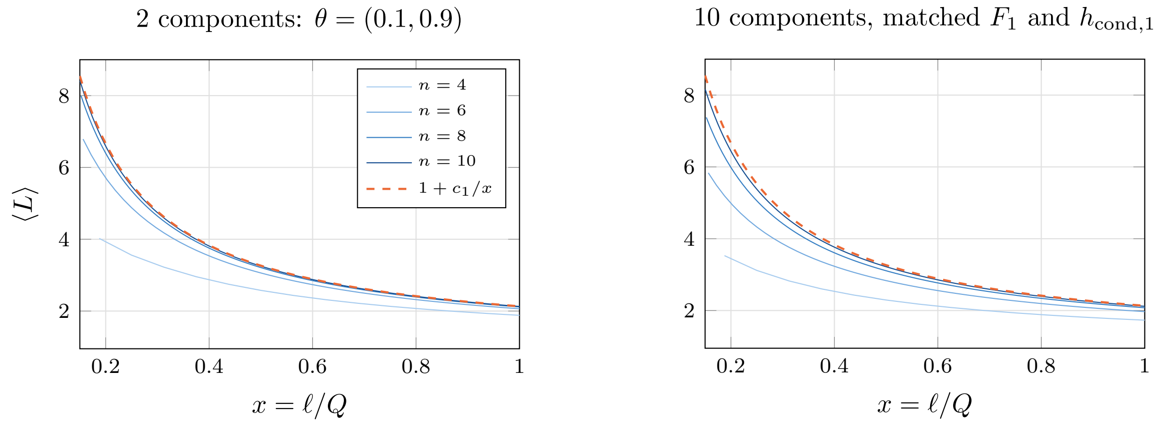


Figure 18: Two binary mixtures with the *same* thermodynamic curve and visibly different finite-size corrections. The right panel uses ten equally weighted biases 0.005, 0.04, 0.1, 0.2, 0.288171 and their complements, the fifth fitted so that $\sum h_2 \simeq 5h_2(0.1)$; complement symmetry fixes the mean bias at $\frac{1}{2}$. Both therefore share $h_0 = 1$ and $h_{\mathrm{cond}} = h_2(0.1)$, hence the same dashed hyperbola, while their finite curves differ because I_k in (96) depends on the whole collection of weights and separations. Panels begin at $x = 0.15$ so the common $1/x$ singularity does not dominate the scale.

7.3.3 What changes when $m > 1$

For $m > 1$ the categorical construction (93) is the general model, and the shared-scalar coin prior is a structured subfamily of it rather than a separate alternative. Use the same Hamming-weight notation at both scales, $|a| := \sum_{b=1}^{m} a_b$ for one m-bit answer and $|\phi_j| := \sum_i |\phi_j(q_i)|$ for the complete table; the signed bias of Chapter 6 is not an independent statistic, being $B_j = 2|\phi_j| - mQ$. Setting $\nu_r(a) = \theta_r^{|a|}(1-\theta_r)^{m-|a|}$ draws one component once for the whole map and makes all mQ table bits independent Bernoulli variables with bias θ_r given $Z = r$, so

$$p_j = \sum_r w_r\, \theta_r^{|\phi_j|}(1-\theta_r)^{mQ-|\phi_j|}. \tag{104}$$

After the shared Z is averaged out, distinct bits are generally correlated both within one answer and across questions, which is why the entropy of one m-bit answer is not $m\,h_2(\sum_r w_r\theta_r)$. Writing

$$F_s := -\sum_{k=0}^{s} \binom{s}{k} P_{s,k} \log_2 P_{s,k} \tag{105}$$

for the entropy of the complete s-bit string, and using $H(\nu_r) = m\,h_2(\theta_r)$, the categorical formulas specialize exactly to $G_\ell = F_{m\ell}$, $h_0 = F_m$ and $h_{\mathrm{cond},m} = m\sum_r w_r h_2(\theta_r)$. The exact finite leverage plotted below is therefore

$$\langle L_{n,\ell}\rangle = \frac{QF_m - (Q-\ell)\big(F_{m(\ell+1)} - F_{m\ell}\big)}{F_{m\ell}}, \qquad \langle L_{n,Q}\rangle = \frac{QF_m}{F_{mQ}}, \tag{106}$$

and since the latent contributes only subextensive information, $F_s/s \to \sum_r w_r h_2(\theta_r)$, giving

$$\boxed{L_m(x) = 1 + \frac{c_m}{x}, \qquad c_m = \frac{F_m - h_{\mathrm{cond},m}}{h_{\mathrm{cond},m}}} \tag{107}$$

provided $h_{\mathrm{cond},m} > 0$. For the mixture above, $F_1 = 1$, $F_2 = 1.680077$, $h_{\mathrm{cond},1} = 0.468996$ and $h_{\mathrm{cond},2} = 0.937991$, so $c_1 = 1.132216$ and $c_2 = 0.791144$. The $m = 2$ panel must therefore use F_2 for the initial answer entropy, not $2h_2(1/2) = 2$.

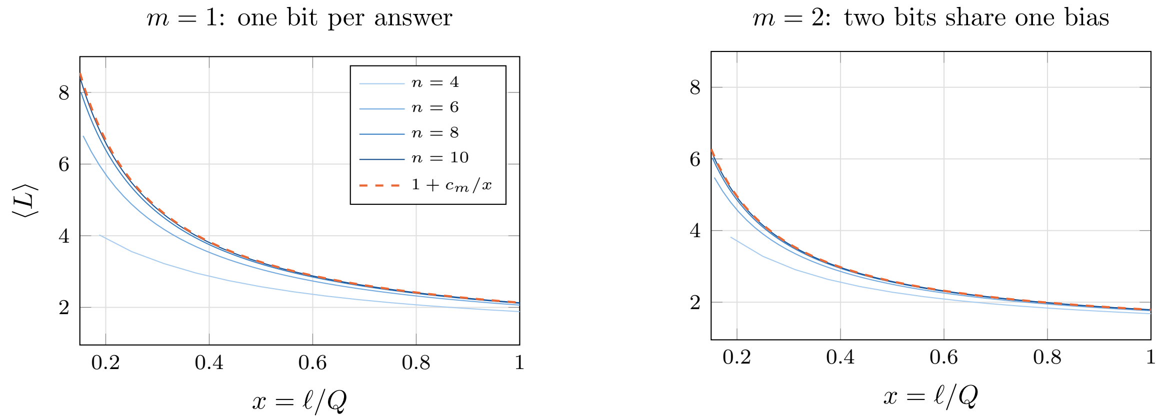


Figure 19: The same shared-scalar latent at two answer widths. Exact blocks use $G_\ell = F_{m\ell}$ and the dashes use (107). The curves approach their hyperbolas at every fixed $x > 0$; no uniform convergence at the singular origin is claimed.

Three superficially similar models are worth separating. With *one global Z shared by every bit*, the construction used here, $G_\ell = F_{m\ell}$ and $h_0 = F_m$. With *one independent global component per output coordinate*, $G_\ell = mF_\ell$, so both initial and residual entropies scale by m and the coefficient stays c_1. With *a fresh component redrawn at every question*, the answers are i.i.d. with distribution ν_{mix}, so $G_\ell = \ell H(\nu_{\mathrm{mix}})$ and $L \equiv 1$: there is no persistent latent variable to learn.

7.3.4 What "two coins are enough" means

For binary answers, a projectively consistent infinitely exchangeable prior is a mixture of i.i.d. Bernoulli laws by de Finetti's theorem. Projective consistency matters: an arbitrary exchangeable prior at each finite n, or an arbitrary sequence of such priors, need not come from one directing measure and need not converge. For a fixed directing measure the macroscopic curve retains only the two entropy summaries $h_0 = h_2(\mathbb{E}\Theta)$ and $h_{\mathrm{cond}} = \mathbb{E}[h_2(\Theta)]$, and after taking a dimensionless ratio only $c = (h_0 - h_{\mathrm{cond}})/h_{\mathrm{cond}}$. For a possibly continuous directing measure the empirical frequency identifies the directing parameter almost surely, so $H(\psi(q_i) \mid \psi(\mathcal{T})) \downarrow \mathbb{E}[h_2(\Theta)]$ and $H(\psi(\mathcal{T}))/k \to h_{\mathrm{cond}}$; the finite-component bound on I_k is not needed for this extension.

Two Bernoulli components realize every finite value of $c \geq 0$. At fixed mean $0 < \mu < 1$ take weights $(1-\mu, \mu)$ and biases $\theta_1 = (1-\xi)\mu$, $\theta_2 = (1-\xi)\mu + \xi$, whose mean remains μ; as ξ moves from 0 toward 1 their mean conditional entropy moves continuously from $h_2(\mu)$ toward 0, so every positive intermediate value is attained. The endpoint $\xi = 1$ has zero entropy density and is excluded from the finite thermodynamic law. This does *not* mean that two atoms reproduce an arbitrary exchangeable prior, its finite-n corrections, or its posterior concentration rate; for $A > 2$, de Finetti gives mixtures of i.i.d. categorical distributions and two atoms do not in general reproduce every marginal-and-entropy pair. The precise conclusion is narrower: every fixed, projectively consistent exchangeable binary prior with positive conditional entropy density has a hyperbolic macroscopic curve; that curve contains only two entropy summaries; and a two-coin model realizes every finite value of their scale-free ratio, one effective coin being the trivial case $L \equiv 1$.

7.4 Non-hyperbolic solvable priors

A stable, projectively consistent exchangeable prior cannot produce a smooth non-hyperbolic macroscopic curve: de Finetti reduces it to an i.i.d. mixture and the preceding section applies. To obtain a profile that changes throughout $0 < x < 1$, the prior must retain structure tied to particular groups or addresses of questions, and a fixed partition into local blocks is the simplest solvable construction. The statement needs its consistency qualification: a separate finite exchangeable law can be chosen at every n without forming one infinite exchangeable family, and such a sequence need be neither an i.i.d. mixture nor possessed of any thermodynamic limit.

7.4.1 Independent blocks of questions

Partition $\mathcal{Q}$ into independent blocks drawn from a fixed finite set of types, writing $\mathcal{P}_n$ for the partition and indexing types by κ. At finite n let $K_{\kappa,n}$ count the type-κ blocks and set $w_{\kappa,n} := r_\kappa K_{\kappa,n}/Q$, the fraction of all questions belonging to them, assuming $w_{\kappa,n} \to w_\kappa$ with $\sum_\kappa w_\kappa = 1$. A type κ carries a size r_κ, a fixed joint distribution on its r_κ answers shared by every block of that type up to relabelling, and a mean subset-entropy profile defined as follows. Let $\mathcal{V} \subseteq \mathcal{Q}$ be one particular block of type κ, fix an ordering of it once and for all, and give every subset $\mathcal{J} \subseteq \mathcal{V}$ the inherited order, so that $\psi(\mathcal{J}) \in \mathcal{A}^{|\mathcal{J}|}$ is its random answer vector. For $j \in \{0, \dots, r_\kappa\}$ define

$$H_\kappa(j) := \binom{r_\kappa}{j}^{-1} \sum_{\substack{\mathcal{J}\subseteq\mathcal{V} \\ |\mathcal{J}|=j}} H\big(\psi(\mathcal{J})\big). \tag{108}$$

Here j is the *number of questions selected from the block*, not a question label, and $H(\psi(\mathcal{J}))$ measures the learner's uncertainty in their *joint answers*, not uncertainty about which questions were selected: the set $\mathcal{J}$ is known. No symmetry is required for this averaged definition, although every zoo member below has the stronger property of **entropy homogeneity**, meaning that $H(\psi(\mathcal{J}))$ itself depends only on $|\mathcal{J}|$. Full permutation invariance is sufficient for that but not necessary: MDS code blocks are entropy homogeneous even though their coordinate distribution is not invariant under every permutation.

The profile records how much answer uncertainty remains visible after looking at j locations inside a block, and its purpose is that successive differences have an operational meaning. Averaging the chain rule over a uniform i-subset $\mathcal{J}$ and then a fresh question drawn uniformly from $\mathcal{V} \setminus \mathcal{J}$ gives the within-block increments

$$\eta_\kappa(i) := H_\kappa(i+1) - H_\kappa(i) = \mathbb{E}_{\mathcal{J},q}\big[H\big(\psi(q) \mid \psi(\mathcal{J})\big)\big], \qquad 0 \le i \le r_\kappa - 1, \tag{109}$$

the mean uncertainty in one fresh answer after i of its block partners have been answered. This is why the block-entropy profile is useful: it converts a joint law on an entire block into exactly the conditional entropies the learning curve needs. The endpoints make the meaning concrete, $H_\kappa(0) = 0$ while $H_\kappa(r_\kappa) = H(\psi(\mathcal{V}))$ is the entropy of the whole block taken jointly. Two independent fair binary answers have profile $(0, 1, 2)$ and increments $(1, 1)$; the same fair bit copied into both questions has profile $(0, 1, 1)$ and increments $(1, 0)$, since after either answer is known the other carries no remaining uncertainty.

For a fresh question in a block of size r_κ, the number of its $r_\kappa - 1$ partners already present in a uniformly random ℓ-set is hypergeometric, so exactly at finite Q, for $0 \le \ell < Q$,

$$\gamma_{n,\ell} = \sum_\kappa w_{\kappa,n} \sum_{i=0}^{r_\kappa-1} \frac{\binom{r_\kappa-1}{i}\binom{Q-r_\kappa}{\ell-i}}{\binom{Q-1}{\ell}}\, \eta_\kappa(i). \tag{110}$$

Holding every r_κ fixed, taking $\ell = \lfloor xQ \rfloor$ and assuming $w_{\kappa,n} \to w_\kappa$, sampling without replacement approaches independent sampling,

$$\frac{\binom{r-1}{i}\binom{Q-r}{\ell-i}}{\binom{Q-1}{\ell}} \longrightarrow \underbrace{\binom{r-1}{i} x^i (1-x)^{r-1-i}}_{\beta_{r-1,i}(x)}, \tag{111}$$

which simultaneously defines the Bernstein basis polynomial $\beta_{r-1,i}$ and, probabilistically, is the limiting probability that exactly i of the fresh question's $r-1$ partners have been asked by macroscopic time x. The limiting profile is therefore

$$\boxed{\gamma(x) = \sum_\kappa w_\kappa \sum_{i=0}^{r_\kappa - 1} \beta_{r_\kappa-1,i}(x)\, \eta_\kappa(i).} \tag{112}$$

At the start of the run no partner has been asked, so only the $i = 0$ term survives and $h_0 = \gamma(0) = \sum_\kappa w_\kappa \eta_\kappa(0) = \sum_\kappa w_\kappa H_\kappa(1)$: the same block-entropy profile supplies both the initial marginal entropy and the entire macroscopic conditional-entropy curve.

To construct a sequence with prescribed fractions, choose integer block counts with $r_\kappa K_{\kappa,n}/Q \to w_\kappa$ and $\sum_\kappa r_\kappa K_{\kappa,n} \le Q$, assigning only $o(Q)$ leftover sites, whose treatment does not affect the limit; at finite n any leftovers can be recorded as an additional block type, so (110) still includes every question and remains exact. If block sizes grow with n the proof changes, a useful sufficient condition for the hypergeometric-to-binomial approximation being $\big(\max_{\mathcal{V}\in\mathcal{P}_n} |\mathcal{V}|\big)^2/Q \to 0$. For countably many types, pointwise frequency convergence must be replaced by an ℓ^1 or tightness condition.

7.4.2 Bernstein polynomials as a profile language

For a single block type, suppressing the type label, equation (112) reads

$$\gamma(x) = \eta(0)(1-x)^2 + 2\eta(1)x(1-x) + \eta(2)x^2 \tag{113}$$

at $r = 3$, and $\eta(0)(1-x)^3 + 3\eta(1)x(1-x)^2 + 3\eta(2)x^2(1-x) + \eta(3)x^3$ at $r = 4$. Some useful coefficient vectors, with h_{fresh} the entropy of one independent fresh answer and h_{clique} the entropy of the value copied through a clique:

block	$(\eta(0), \dots, \eta(r-1))$	profile
independent fresh answers	$(h_{\text{fresh}}, \dots, h_{\text{fresh}})$	h_{fresh}
clique	$(h_{\text{clique}}, 0, \dots, 0)$	$h_{\text{clique}}(1-x)^{r-1}$
parity	$(m, \dots, m, 0)$	$m(1-x^{r-1})$
(r,k) MDS	k entries m, then zeros	$m \Pr[\mathrm{Bin}(r-1,x) \le k-1]$

Entropy submodularity gives $m \ge \eta_\kappa(0) \ge \cdots \ge \eta_\kappa(r_\kappa - 1) \ge 0$, and the derivative identity for Bernstein polynomials makes the macroscopic monotonicity explicit,

$$\gamma'(x) = \sum_\kappa w_\kappa (r_\kappa - 1) \sum_{i=0}^{r_\kappa-2} \big(\eta_\kappa(i+1) - \eta_\kappa(i)\big)\beta_{r_\kappa-2,i}(x) \;\le\; 0, \tag{114}$$

so γ is nonincreasing as the general entropy calculus requires. Leverage itself need not be monotone.

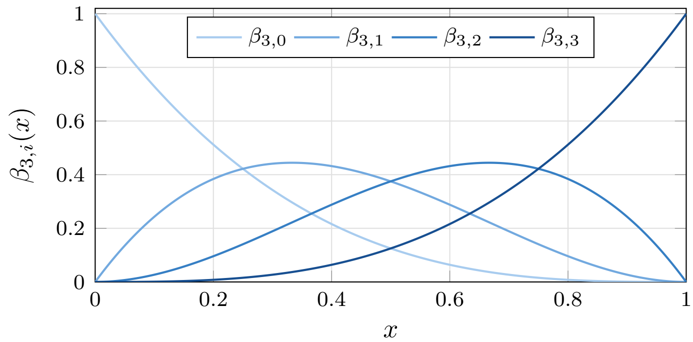


Figure 20: The degree-three Bernstein basis: bump functions scanning left to right, the i-th peaked at $i/3$. A block profile is a positive combination of such bumps weighted by the microscopic conditional entropies $\eta(i)$, so the shape of L becomes a question about *where the coefficients drop.*

Equation (112) describes every prior in the stated class of independent blocks drawn from a fixed finite list of bounded types, and mixing types mixes their profiles linearly. There is also a useful approximation result when the answer alphabet is allowed to grow: mixtures of common-length MDS blocks realize any nonincreasing Bernstein coefficient sequence, and Bernstein polynomials approximate every continuous nonincreasing profile. A Reed–Solomon block of length r needs $A \geq r$, however, so sending $r \to \infty$ in that approximation also requires $m \to \infty$, or a different family of realizable entropy vectors. It is not a completeness theorem at fixed answer width.

7.4.3 Why leverage can have an interior maximum

The destroyed-entropy density is $h_0 - (1-x)\gamma(x)$, so after subtracting the received density the cumulative deduced density is $h_0 - (1-x)\gamma(x) - g(x)$, whose derivative, for differentiable γ, is

$$\frac{d}{dx}\Big[h_0 - (1-x)\gamma(x) - g(x)\Big] = -(1-x)\gamma'(x) \;\geq\; 0. \tag{115}$$

Hence $-\gamma'(x)$, weighted by the unasked fraction, specifies when stored information is released. Since $L(x) = 1 + \big[h_0 - (1-x)\gamma(x) - g(x)\big]/g(x)$,

$$L'(x) = \frac{-(1-x)\gamma'(x)g(x) - \big[h_0 - (1-x)\gamma(x) - g(x)\big]\gamma(x)}{g(x)^2}, \tag{116}$$

so leverage rises when the current deduction-to-reception rate exceeds its accumulated average and falls when it drops below it. For an explicit example mix a fraction w of $(r,k) = (4,2)$ MDS questions with a fraction $1-w$ of fresh questions, giving

$$\frac{\gamma(x)}{m} = 1 - 3wx^2 + 2wx^3, \qquad \frac{g(x)}{m} = x - wx^3 + \frac{w}{2}x^4, \tag{117}$$

and cumulative deduced density $3wx^2 - 4wx^3 + \frac{3w}{2}x^4$ per bit. Consequently $L(x) = 1 + 3wx + O(x^2)$ as $x \downarrow 0$, whereas direct differentiation at the other endpoint gives

$$L'(1) = -\frac{w}{2}\,\frac{1-w}{(1-w/2)^2} \;<\; 0 \qquad (0 < w < 1). \tag{118}$$

Thus $L'(0^+) = 3w > 0$ while $L'(1) < 0$, and continuity forces an interior maximum. The peak is produced by a threshold release followed by continued ordinary reception.

7.5 An exotic zoo of solvable thermodynamic priors

The following priors are designed to release stored information at different stages of the run. Their profiles are explicit, so their leverage follows from (91). Full derivations appear in Appendices A–F.

7.5.1 Cliques: constant leverage

For a clique block $\mathcal{V} \in \mathcal{P}_n$ of size r, draw one shared value $Z_\mathcal{V} \sim \nu_\mathcal{V}$ and copy it to all questions in $\mathcal{V}$. With $h_\mathcal{V} := H(\nu_\mathcal{V})$ we have $H_\mathcal{V}(0) = 0$ and $H_\mathcal{V}(j) = h_\mathcal{V}$ for $j \geq 1$, so the increments are $(h_\mathcal{V}, 0, \ldots, 0)$. For a common block size, letting h_{clique} denote the question-weighted mean of the $h_\mathcal{V}$,

$$\gamma(x) = h_{\text{clique}}(1-x)^{r-1}, \qquad g(x) = \frac{h_{\text{clique}}}{r}\big[1-(1-x)^r\big], \qquad \boxed{L(x) \equiv r.} \tag{119}$$

The result is exact at finite n for an exact clique partition, or with only deterministic zero-entropy leftovers. Mixing clique sizes at a common entropy scale does not interpolate between constants: the initial leverage is the arithmetic mean of the sizes while the final leverage is their harmonic mean.

7.5.2 Parity: deductions near the end

For a binary block $\mathcal{V}$ of size r, fix a coset label $\sigma_\mathcal{V} \in \{0,1\}$ and choose its answer tuple uniformly from that parity coset,

$$\Pr\big(\psi(\mathcal{V}) = y_\mathcal{V}\big) = 2^{-(r-1)}\,\mathbf{1}\Big[\textstyle\bigoplus_{q\in\mathcal{V}} y_q = \sigma_\mathcal{V}\Big]. \tag{120}$$

Every proper subset is uniform, so $\eta(i) = 1$ for $0 \leq i \leq r-2$ and $\eta(r-1) = 0$, giving $\gamma(x) = 1 - x^{r-1}$, $g(x) = x - x^r/r$ and

$$\boxed{L(x) = \frac{x + (1-x)x^{r-1}}{x - x^r/r}.} \tag{121}$$

For $r \geq 3$ this rises from 1 to $r/(r-1)$. The case $r = 2$ is exceptional: it is an equality or inequality clique, with $L \equiv 2$. The derivative of the cumulative deduced density, $-(1-x)\gamma'(x) = (r-1)x^{r-2}(1-x)$, concentrates late in the run.

7.5.3 Tilted parity

Uniform parity is not the only solvable law on a parity coset. For $0 < \theta < 1$ draw i.i.d. Bernoulli(θ) bits and condition their parity to be σ. With $\varepsilon := 1 - 2\theta$ and $\zeta_{r,\sigma} = \frac{1}{2}\big(1 + (-1)^\sigma \varepsilon^r\big)$ the block law is

$$\Pr_\sigma\big(\psi(\mathcal{V}) = y_\mathcal{V}\big) = \frac{\theta^{|y_\mathcal{V}|}(1-\theta)^{r-|y_\mathcal{V}|}\mathbf{1}[\oplus_q y_q = \sigma]}{\zeta_{r,\sigma}}, \tag{122}$$

and if v bits remain unseen with required parity σ_{rem}, the next bit is 1 with probability

$$P_{\text{next}}(v, \sigma_{\text{rem}}) = \frac{\theta\big[1 - (-1)^{\sigma_{\text{rem}}}\varepsilon^{\,v-1}\big]}{1 + (-1)^{\sigma_{\text{rem}}}\varepsilon^{\,v}}. \tag{123}$$

A finite average of $h_2(P_{\text{next}})$ gives each $\eta(i)$, the exact weights being derived in Appendix C. The final member remains determined, so $\eta(r-1) = 0$, while earlier coefficients now decrease gradually rather than staying exactly flat. At $r = 4$, $\theta = 0.3$, $\sigma = 1$,

$$\eta = (0.912441,\ 0.898876,\ 0.811317,\ 0), \tag{124}$$

which gives $L(0^+) \simeq 1.0446$ and $L(1) \simeq 1.3915$. The parameter ε controls how far the profile bends away from uniform parity.

7.5.4 MDS blocks: a movable threshold

Let answers be symbols of $\mathbb{F}_A$ and let a block $\mathcal{V}$ of size r be uniform on an $[r,k]_A$ MDS code, a Reed–Solomon realization using r distinct field points and therefore requiring $A \geq r$. Any k answers determine the whole block while any $j \leq k$ answers are uniform, so $H_{\mathrm{MDS}}(j) = m\min(j,k)$ and $\eta(i) = m$ for $i < k$, zero afterwards. Writing $\tau_{r,k}(x) := \Pr[\mathrm{Bin}(r-1,x) \leq k-1]$ we obtain $\gamma(x) = m\,\tau_{r,k}(x)$, and the beta integral gives

$$\rho := \frac{k}{r}, \qquad \int_0^1 \tau_{r,k}(x)\,dx = \rho, \qquad \text{hence} \qquad L(1) = \frac{1}{\rho}. \tag{125}$$

If $n \to \infty$ is taken first and then $r \to \infty$ with $k/r \to \rho$, the profile approaches a step at $x = \rho$, the answer width having to grow for such blocks to exist. To produce a peak, mix a code fraction w with a fresh fraction $1-w$:

$$\frac{\gamma(x)}{m} = w\,\tau_{r,k}(x) + (1-w), \qquad \boxed{L(x) = \frac{w\big[1-(1-x)\tau_{r,k}(x)\big] + (1-w)x}{w\int_0^x \tau_{r,k}(t)\,dt + (1-w)x}.} \tag{126}$$

The code releases deductions around its threshold while fresh questions continue to add received information afterwards, making the cumulative curve decline past its maximum.

7.5.5 Local noise and full support

The hard clique, parity and code priors assign probability zero to many maps. For $0 \leq \delta \leq 1$ a general local lapse replaces the hard law $P_{\mathcal{V}}$ on a block by $P_{\mathcal{V},\delta} = (1-\delta)P_{\mathcal{V}} + \delta A^{-|\mathcal{V}|}$, so that term by term the map prior becomes

$$\boxed{p_{\delta,j} = \prod_{\mathcal{V}\in\mathcal{P}_n} \Big[(1-\delta_{\mathcal{V},n})P_{\mathcal{V}}\big(\phi_j(\mathcal{V})\big) + \delta_{\mathcal{V},n}A^{-|\mathcal{V}|}\Big].} \tag{127}$$

The map prior has full support whenever every $\delta_{\mathcal{V},n} > 0$, and the lapse is local: blocks remain independent, so only their entropy coefficients change. For fixed block sizes entropy is continuous on the finite probability simplex, so $\eta_{\mathcal{V},\delta}(i) \to \eta_{\mathcal{V},0}(i)$ as $\delta \to 0$, and if $\max_{\mathcal{V}} \delta_{\mathcal{V},n} \to 0$ the hard and noisy priors share a thermodynamic profile. More generally it is enough that the question-weighted coefficient perturbation vanish,

$$\sum_{\mathcal{V}\in\mathcal{P}_n} \frac{|\mathcal{V}|}{Q} \max_i \big|\eta_{\mathcal{V},\delta_{\mathcal{V},n}}(i) - \eta_{\mathcal{V},0}(i)\big| \;\longrightarrow\; 0. \tag{128}$$

At fixed $\delta > 0$ the noise does *not* vanish merely because $n \to \infty$; it produces a solvable but displaced profile. If block sizes also grow, $\delta_n \to 0$ alone may be insufficient: for noisy parity blocks of size $r(n)$ one needs roughly $r(n)\delta_n \to 0$. Two coefficient changes are especially simple. Uniform parity followed by independent bit flips at rate δ keeps $\eta(i) = 1$ for $i \leq r-2$ and lifts only the cliff,

$$\eta(r-1) = h_2\!\left(\frac{1-(1-2\delta)^r}{2}\right), \tag{129}$$

while an MDS block lapsed to the uniform block distribution keeps $H_\delta(j) = jm$ below threshold and turns the zero coefficients above it into a positive tail, with the exact entropy given in Appendix D. Merely giving the shared clique value full alphabet support is not enough, since the block would still contain only constant strings: independent output noise or a lapse is required for full map support.

7.5.6 Arbitrary mixtures and the cocktail

Suppose component type κ occupies a fraction w_κ of questions and has initial marginal entropy $h_{0,\kappa}$, profile γ_κ, integral g_κ and leverage L_κ. Independence across disjoint question sets gives $h_0 = \sum_\kappa w_\kappa h_{0,\kappa}$, $\gamma = \sum_\kappa w_\kappa \gamma_\kappa$ and $g = \sum_\kappa w_\kappa g_\kappa$, and therefore

$$\boxed{L(x) = \frac{\sum_\kappa w_\kappa g_\kappa(x) L_\kappa(x)}{\sum_\kappa w_\kappa g_\kappa(x)}.} \tag{130}$$

Leverage is a received-entropy-weighted average of component leverages, not a simple question-fraction average, and (130) is a symbolic solution for any mixture whose component profiles are known.

As a worked cocktail take $m = 4$, with one quarter of the questions following the two-coin mixture $\theta \in \{0.2, 0.7\}$ with weights $(\frac{1}{4}, \frac{3}{4})$ and one scalar latent bias shared by all its bits; one half belonging to lapsed $(r, k) = (16, 4)$ Reed–Solomon blocks with lapse $\delta = 0.1$; and one quarter fresh with independent bit bias 0.3. Normalizing entropy per output bit and writing $h_{\text{cocktail}} := h_0/m$, $\gamma_{\text{cocktail}} := \gamma/m$ and $g_{\text{cocktail}} := g/m$, the coin component contributes $F_4/4 = 0.94828746$ initially and $h_{\text{cond,coin}} = 0.84145020$ afterwards, so

$$h_{\text{cocktail}} = \tfrac{1}{4}(0.94828746) + \tfrac{1}{2} + \tfrac{1}{4}h_2(0.3) = 0.95739459, \tag{131}$$

while for $x > 0$, with γ_{lapse} the lapsed-code profile per bit,

$$\gamma_{\text{cocktail}}(x) = \tfrac{1}{4}h_{\text{cond,coin}} + \tfrac{1}{2}\gamma_{\text{lapse}}(x) + \tfrac{1}{4}h_2(0.3), \qquad \int_0^1 \gamma_{\text{lapse}} = 0.33232806. \tag{132}$$

Substituting into (91) gives $L = \big[h_{\text{cocktail}} - (1-x)\gamma_{\text{cocktail}}\big]/g_{\text{cocktail}}$, whose initial divergence has coefficient $\lim_{x\downarrow 0} xL(x) = \big[h_{\text{cocktail}} - \gamma_{\text{cocktail}}(0^+)\big]/\gamma_{\text{cocktail}}(0^+) \simeq 0.02870$, a dip of approximately 1.499 at $x \simeq 0.0836$, a peak of approximately 2.114 at $x \simeq 0.350$, and $L(1) \simeq 1.60408$. The initial hyperbolic decline comes from the global coin latent, the middle peak from the code threshold, and the final decline from fresh and lapsed entropy received after the code deductions have mostly been released.

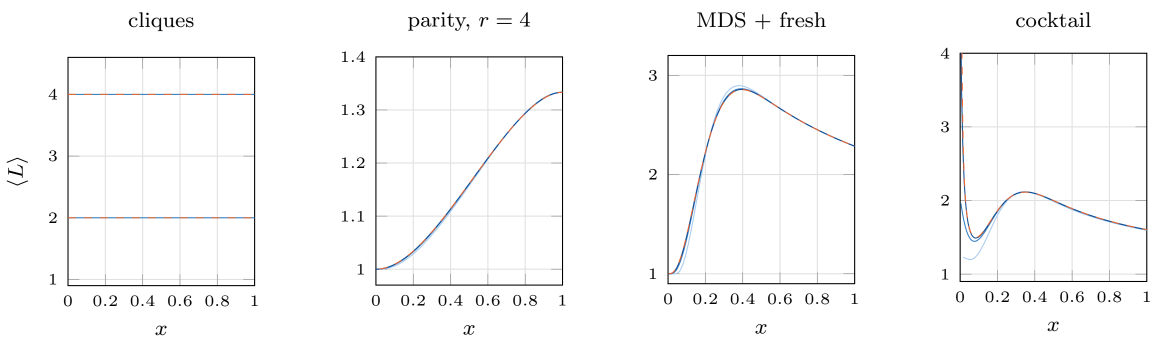


Figure 21: Finite learning curves at $n = 6, 8, 10$ (light to dark) against their dashed thermodynamic limits. Cliques of size 2 and 4 are exactly flat at every finite size; parity rises to $r/(r-1)$; the diluted code peaks near its threshold; and the cocktail declines, dips, peaks and declines again. In the cocktail panel the dashed curve uses the shared-latent initial entropy $F_4/4$ and equations (131)–(132).

7.5.7 The zoo side by side

member	microscopic rule	limiting profile	leverage shape
coin mixture	a global latent selects an i.i.d. component	constant on $x > 0$, jump at 0	$1 + c/x$
clique	one value copied through each block	$h_{\text{clique}}(1-x)^{r-1}$	exactly r
parity	one final parity constraint	$m(1 - x^{r-1})$	rising for $r \geq 3$
tilted parity	biased product law conditioned on parity	(112) with (124)	tilt-dependent
MDS	any k answers determine an r-block	$m\,\tau_{r,k}(x)$	threshold rise
MDS + fresh	threshold block plus ordinary sites	$m[w\tau_{r,k} + (1-w)]$	interior peak
cocktail	coin, lapsed code and fresh	weighted sum	decline, dip, peal

The shape of a well-measured learning curve can therefore provide evidence about the **timing** of usable correlations: global and front-loaded, first-touch, last-touch, or threshold-like. It does not identify a unique microscopic prior. Different priors can share the same entropy profile, and an actual trained model may be misspecified, non-Bayesian, or imperfectly calibrated. The central scale distinction is now explicit: the $n \to \infty$ curve sees only extensive-time learning, and any structure learned in $O(1)$ or $o(Q)$ questions collapses into a singularity or jump at $x = 0$. For Bayesian priors this means that a finite-dimensional global latent may profoundly affect early learning yet disappear from the interior profile. To obtain nontrivial behaviour throughout $0 < x < 1$, the prior must distribute uncertainty and correlations across $\Theta(Q)$ local degrees of freedom. Macroscopic leverage classifies when stored information becomes available, not every detail of how the prior represents it.

Appendices A–F derive the block formulas of Chapter 7 in a common order: specify the probability law inside one block; compute its subset entropies $H_\kappa(j)$; take differences $\eta_\kappa(i) = H_\kappa(i+1) - H_\kappa(i)$; insert those differences into the Bernstein profile; integrate the profile and substitute it into the leverage law. Appendix G discusses the inverse problem and Appendix H collects endpoint and boundary-layer consequences of the general limit.

A The general block calculation

A.1 Exact mean block entropy at finite n

Let $\mathcal{V} \in \mathcal{P}_n$ be a block and define the entropy averaged over all j-subsets of it,

$$H_{\mathcal{V}}(j) := \binom{|\mathcal{V}|}{j}^{-1} \sum_{\substack{\mathcal{J}\subseteq\mathcal{V}\\ |\mathcal{J}|=j}} H\big(\psi(\mathcal{J})\big), \tag{133}$$

every term of which is equal when the block is entropy homogeneous. If $\mathcal{T}$ is a uniformly random ℓ-subset of $\mathcal{Q}$ then its intersection count with $\mathcal{V}$ is hypergeometric,

$$\Pr(|\mathcal{T} \cap \mathcal{V}| = j) = \frac{\binom{|\mathcal{V}|}{j}\binom{Q-|\mathcal{V}|}{\ell-j}}{\binom{Q}{\ell}}. \tag{134}$$

Different blocks have dependent intersection counts, since those counts sum to ℓ, but entropy adds across independent blocks and expectation is linear, so

$$\boxed{G_{n,\ell} = \sum_{\mathcal{V}\in\mathcal{P}_n} \sum_{j=0}^{|\mathcal{V}|} \frac{\binom{|\mathcal{V}|}{j}\binom{Q-|\mathcal{V}|}{\ell-j}}{\binom{Q}{\ell}} H_{\mathcal{V}}(j)} \tag{135}$$

exactly, with no large-n approximation anywhere.

A.2 Exact conditional-entropy increments

Choose a fresh question q uniformly and then ℓ of the remaining $Q-1$ questions. Conditional on q belonging to a type-κ block of size r_κ, the probability of seeing exactly i already-asked partners is $\binom{r_\kappa-1}{i}\binom{Q-r_\kappa}{\ell-i}/\binom{Q-1}{\ell}$, and only those partners affect the fresh answer because blocks are independent. Choosing an i-subset and then a fresh coordinate uniformly shows that $\eta_\kappa(i) = H_\kappa(i+1) - H_\kappa(i)$ is the corresponding mean conditional entropy even without entropy homogeneity, so for $0 \le \ell < Q$,

$$\boxed{\gamma_{n,\ell} = G_{n,\ell+1} - G_{n,\ell} = \sum_\kappa w_{\kappa,n} \sum_{i=0}^{r_\kappa-1} \frac{\binom{r_\kappa-1}{i}\binom{Q-r_\kappa}{\ell-i}}{\binom{Q-1}{\ell}}\, \eta_\kappa(i).} \tag{136}$$

A.3 Hypergeometric to binomial

Fix r and i, set $\ell = \lfloor xQ \rfloor$, and expose a particular set of i partners that are selected and $r-1-i$ that are not. Its probability is a product of $r-1$ factors of the form $(\ell+O(r))/(Q+O(r))$ or $(Q-\ell+O(r))/(Q+O(r))$; each selected factor tends to x and each unselected factor to $1-x$. There are $\binom{r-1}{i}$ such choices, so $\Pr(\text{exactly } i \text{ partners}) \to \binom{r-1}{i}x^i(1-x)^{r-1-i}$, with total-variation error $O(r^2/Q)$ at fixed r. Thus (136) tends to (112).

A.4 Integrating a Bernstein profile

Direct integration of one basis function gives

$$\int_0^x \beta_{r-1,i}(t)\,dt = \binom{r-1}{i}\int_0^x t^i(1-t)^{r-1-i}\,dt, \qquad \int_0^1 \beta_{r-1,i}(t)\,dt = \frac{1}{r}, \tag{137}$$

so that $g(x) = \sum_\kappa w_\kappa \sum_i \eta_\kappa(i)\binom{r_\kappa-1}{i}\int_0^x t^i(1-t)^{r_\kappa-1-i}dt$ and, at the end of the run,

$$g(1) = \sum_\kappa \frac{w_\kappa}{r_\kappa}\sum_i \eta_\kappa(i) = \sum_\kappa \frac{w_\kappa}{r_\kappa} H_\kappa(r_\kappa), \tag{138}$$

exactly the prior entropy per question: there are $w_\kappa Q/r_\kappa$ independent type-κ blocks, each carrying $H_\kappa(r_\kappa)$ bits.

A.5 MDS mixtures as Bernstein coefficient synthesizers

Fix a common block length r over $\mathbb{F}_A$ with $A \ge r$, so that $[r,k]_A$ Reed–Solomon codes exist for $k = 1, \dots, r$, and normalize entropies by m. A dimension-k block contributes coefficient 1 at indices $i < k$ and 0 afterwards, so if dimension k is used with question fraction w_k the mixed normalized coefficients are $\eta_{\text{target}}(i) = \sum_{k=i+1}^r w_k$. Conversely, take any nonincreasing sequence $1 \ge \eta_{\text{target}}(0) \ge \dots \ge \eta_{\text{target}}(r-1) \ge 0$, set $\eta_{\text{target}}(r) = 0$, and choose

$$w_k = \eta_{\text{target}}(k-1) - \eta_{\text{target}}(k) \qquad (1 \le k \le r), \tag{139}$$

which is nonnegative and reproduces the sequence, any slack $1 - \eta_{\text{target}}(0)$ being assigned to frozen zero-entropy blocks. Bernstein's approximation theorem then approximates a continuous nonincreasing normalized target profile by taking $\eta_{\text{target}}(i) = \gamma_{\text{target}}(i/(r-1))$. This establishes a profile-synthesis result only under the accompanying realization conditions, $r^2/Q \to 0$ along a diagonal limit, $A \ge r$, and leftovers of vanishing density; it is not a completeness theorem at fixed m.

B Cliques

B.1 Block law and entropy vector

Let $\mathcal{V} \in \mathcal{P}_n$ have size r. Draw an m-bit shared value $Z_\mathcal{V} \sim \nu_\mathcal{V}$ and set $\psi(q) = Z_\mathcal{V}$ for every $q \in \mathcal{V}$, so that $\Pr(\psi(\mathcal{V}) = y_\mathcal{V}) = \nu_\mathcal{V}(y_{q_1})\mathbf{1}[y_{q_1} = \cdots = y_{q_r}]$. Observing no coordinate has zero entropy; observing at least one reveals $Z_\mathcal{V}$, after which further coordinates add nothing:

$$H_\mathcal{V}(0) = 0, \qquad H_\mathcal{V}(j) = h_\mathcal{V} \ (1 \le j \le r), \qquad \eta_\mathcal{V} = (h_\mathcal{V}, 0, \ldots, 0). \tag{140}$$

B.2 Profile, integral, and leverage

Only the $i = 0$ Bernstein term survives, so $\gamma(x) = h_{\text{clique}}(1-x)^{r-1}$ and, substituting $u = 1-t$, $g(x) = \frac{h_{\text{clique}}}{r}[1 - (1-x)^r]$. The initial marginal entropy is $h_0 = h_{\text{clique}}$ and the destroyed density is

$$h_0 - (1-x)\gamma(x) = h_{\text{clique}}\big[1 - (1-x)^r\big] = r\,g(x), \qquad \text{hence} \qquad L(x) = r. \tag{141}$$

For example, one four-question clique with a fair binary value has prior mass $\frac{1}{2}$ on 0000 and $\frac{1}{2}$ on 1111; its profile is $(1-x)^3$, its received density $[1-(1-x)^4]/4$, and its leverage 4 at every stage.

B.3 Why the result is exact at finite size

A block is learned if and only if it has been touched at least once, and its probability of remaining untouched after a random ℓ-subset is $\binom{Q-r}{\ell}/\binom{Q}{\ell}$. For Q/r equal-entropy blocks the expected received information is $\frac{Q}{r} h_{\text{clique}}\big[1 - \binom{Q-r}{\ell}/\binom{Q}{\ell}\big]$ while the expected destroyed table entropy is $Q\,h_{\text{clique}}\big[1 - \binom{Q-r}{\ell}/\binom{Q}{\ell}\big]$. Their ratio is exactly r at every non-null stage.

B.4 Mixing clique sizes

Suppose a question fraction w_κ lies in size-r_κ cliques with a common entropy scale. Substituting each type's destroyed factor $1 - (1-x)^{r_\kappa}$ and received factor $r_\kappa^{-1}[1-(1-x)^{r_\kappa}]$ gives

$$L(x) = \frac{\sum_\kappa w_\kappa\big[1-(1-x)^{r_\kappa}\big]}{\sum_\kappa (w_\kappa/r_\kappa)\big[1-(1-x)^{r_\kappa}\big]}, \qquad L(0^+) = \sum_\kappa w_\kappa r_\kappa, \qquad L(1) = \Big(\sum_\kappa \frac{w_\kappa}{r_\kappa}\Big)^{-1}, \tag{142}$$

the arithmetic and harmonic means of the sizes. A mixture of flat clique curves is therefore generally a decreasing curve, not another constant. If type-κ cliques instead have mean entropy h_κ, the entropy-weighted endpoints are $L(0^+) = \sum w_\kappa h_\kappa r_\kappa / \sum w_\kappa h_\kappa$ and $L(1) = \sum w_\kappa h_\kappa / \sum w_\kappa h_\kappa / r_\kappa$.

B.5 Noisy cliques

For binary answers draw the shared bit $Z \sim \nu$ and let $\psi(q_j) = Z \oplus E_j$ with E_j i.i.d. Bernoulli(δ). Writing $\mathcal{J}_i := \{q_1, \ldots, q_i\}$, the posterior on the shared value after observing $y_{\mathcal{J}_i}$ is

$$P_i(z \mid y_{\mathcal{J}_i}) \ \propto \ \nu(z) \prod_{j=1}^{i} \big[(1-\delta)\mathbf{1}(y_{q_j} = z) + \delta\mathbf{1}(y_{q_j} \neq z)\big], \tag{143}$$

so the next-one probability is $\delta + (1-2\delta)P_i(1 \mid y_{\mathcal{J}_i})$ and

$$\eta_\delta(i) = \sum_{y_{\mathcal{J}_i} \in \{0,1\}^i} \Pr\big(\psi(\mathcal{J}_i) = y_{\mathcal{J}_i}\big)\, h_2\big(\delta + (1-2\delta)P_i(1 \mid y_{\mathcal{J}_i})\big). \tag{144}$$

For every $0 < \delta < 1$ all binary strings have positive probability, and as $\delta \to 0$ this tends to (140) term by term. At fixed nontrivial noise the exact $L \equiv r > 1$ law is lost, and at $\delta = \frac{1}{2}$ the output is i.i.d. fair and the different flat law $L \equiv 1$ results.

C Uniform, tilted, and noisy parity

C.1 Uniform parity

Fix $\sigma \in \{0,1\}$ and choose the binary answers in $\mathcal{V}$ uniformly subject to $\bigoplus_{q \in \mathcal{V}} \psi(q) = \sigma$. There are 2^{r-1} admissible strings; fixing any $j \le r-1$ coordinates at any values leaves the remaining $r-j$ obeying one parity equation and therefore 2^{r-j-1} completions, independent of the fixed values. Every proper subset is consequently uniform,

$$H_{\mathcal{V}}(j) = j \quad (0 \le j \le r-1), \qquad H_{\mathcal{V}}(r) = r-1, \qquad \eta(i) = \begin{cases} 1, & 0 \le i \le r-2, \\ 0, & i = r-1. \end{cases} \tag{145}$$

For m-bit answers the same formulas acquire a factor m if parity is imposed independently on each bit, or as one additive equation over $\mathbb{F}_A$.

C.2 Profile and leverage

The Bernstein basis sums to one, so removing only the final term gives $\gamma(x) = \sum_{i=0}^{r-2} \beta_{r-1,i}(x) = 1 - x^{r-1}$ and $g(x) = x - x^r/r$. The destroyed density is $1-(1-x)(1-x^{r-1}) = x + (1-x)x^{r-1}$, so L is (121), and for $r \ge 3$ direct differentiation yields

$$L'(x) = \frac{x^{r-1}(1-x)\big[r(r-2) - \sum_{j=1}^{r-2} x^j\big]}{r\big(x - x^r/r\big)^2} \;\ge\; 0, \qquad L(0^+) = 1, \qquad L(1) = \frac{r}{r-1}, \tag{146}$$

with $L \equiv 2$ at $r = 2$. For the smallest genuinely different example take one odd-parity block with $r = 4$, uniform on the eight odd-weight strings, so that $\gamma(x) = 1 - x^3$, $g(x) = x - x^4/4$. The derivative of the cumulative deduced density, $-(1-x)\gamma'(x) = (r-1)x^{r-2}(1-x)$, is proportional to a $\mathrm{Beta}(r-1, 2)$ density; it is also the density of the stage at which a block's penultimate queried member determines its final member, divided by r questions per block.

C.3 Tilted parity: exact subset probabilities

For $0 < \theta < 1$ start with i.i.d. Bernoulli(θ) bits and condition on the parity. With $\varepsilon = 1 - 2\theta$, the probability that v i.i.d. bits have parity σ_{rem} is $\zeta_{v,\sigma_{\mathrm{rem}}} = \frac{1}{2}\big(1 + (-1)^{\sigma_{\mathrm{rem}}}\varepsilon^v\big)$, and the normalized tilted-coset law is (122). Take a particular realized tuple $y_{\mathcal{J}}$ on $\mathcal{J} \subset \mathcal{V}$ with $|\mathcal{J}| = j$ and Hamming weight z; its unobserved coordinates must have parity $\sigma \oplus (z \bmod 2)$, and summing them out gives

$$\Pr_\sigma\big(\psi(\mathcal{J}) = y_{\mathcal{J}}\big) = \theta^z (1-\theta)^{j-z}\, \frac{1 + (-1)^{\sigma \oplus (z \bmod 2)} \varepsilon^{r-j}}{1 + (-1)^\sigma \varepsilon^r} \;=:\; \chi_\sigma(j,z). \tag{147}$$

The subset entropy is then the finite sum $H_\sigma(j) = -\sum_{z=0}^{j} \binom{j}{z} \chi_\sigma(j,z) \log_2 \chi_\sigma(j,z)$, and the coefficients are exactly $\eta(i) = H_\sigma(i+1) - H_\sigma(i)$.

C.4 Tilted parity: predictive form

The same result can be organized as a conditional prediction. Suppose $|\mathcal{J}| = i$ questions have been seen, so $v := r - i$ bits remain; if their required parity is σ_{rem} then $P_{\text{next}}(v, \sigma_{\text{rem}})$ is (123). Letting σ_{seen} be the parity of the i seen bits, conditional on total parity σ,

$$\Pr(\sigma_{\text{seen}} = z \mid \sigma) = \frac{\left[1 + (-1)^z \varepsilon^i\right]\left[1 + (-1)^{\sigma \oplus z} \varepsilon^v\right]}{2\left[1 + (-1)^\sigma \varepsilon^r\right]}, \qquad \boxed{\eta(i) = \sum_{z=0}^{1} \Pr(\sigma_{\text{seen}} = z \mid \sigma)\, h_2\big(P_{\text{next}}(v, \sigma \oplus z)\big).} \tag{148}$$

When $\theta = \frac{1}{2}$ we have $\varepsilon = 0$ and this reduces to (145); when $v = 1$ the remaining bit is fixed and $\eta(r-1) = 0$. At $r = 4$, $\theta = 0.3$, $\sigma = 1$, evaluation gives (124).

C.5 Uniform parity followed by bit-flip noise

Return to the uniform coset and flip every bit independently with probability δ. The parity changes precisely when an odd number of flips occurs, with $P_{\text{odd}}(r, \delta) = \frac{1}{2}\left[1 - (1-2\delta)^r\right]$, so convolution with the flip channel produces a mixture of the two uniform parity cosets with weights $1 - P_{\text{odd}}$ and P_{odd}. Any $r-1$ coordinates remain uniform, so $\eta_\delta(i) = 1$ for $i \le r-2$, while given $r-1$ coordinates the uncertainty in the final bit is exactly the uncertainty in the resulting parity label:

$$\boxed{\gamma_\delta(x) = 1 - \left[1 - h_2(P_{\text{odd}}(r, \delta))\right] x^{r-1}.} \tag{149}$$

For fixed r any $\delta_n \to 0$ recovers the hard profile; if the block size grows one instead needs $P_{\text{odd}}(r(n), \delta_n) \to 0$, which for small noise requires roughly $r(n)\delta_n \to 0$. These formulas rely on a uniform starting coset: tilted parity followed by noise remains solvable by finite sums but is not uniform within each resulting coset.

D MDS and Reed–Solomon blocks

D.1 Block construction

Work over $\mathbb{F}_A$ and choose r distinct evaluation points $\alpha_1, \dots, \alpha_r$, which requires $r \le A$. Draw a polynomial $f(z) = d_0 + d_1 z + \cdots + d_{k-1} z^{k-1}$ uniformly from the A^k polynomials of degree below k and set $\psi(q_j) = f(\alpha_j)$, a uniform Reed–Solomon codeword. Two elementary interpolation facts determine all entropies: values at any k distinct points determine f and hence all r answers; and values at any $j \le k$ points are uniform on $\mathbb{F}_A^j$, since prescribing those j values, choosing values at another $k - j$ points freely and interpolating shows that exactly A^{k-j} polynomials realize every prescription. Therefore

$$H_{\text{MDS}}(j) = m \min(j, k), \qquad \eta(i) = \begin{cases} m, & 0 \le i \le k-1, \\ 0, & k \le i \le r-1. \end{cases} \tag{150}$$

The same entropy law holds for any uniform coset of an $[r, k]_A$ MDS code. Repetition ($k = 1$) and single-parity-check ($k = r - 1$) codes give the clique and parity entropy laws over any alphabet, although for small fields and large r they are not obtained from r distinct Reed–Solomon evaluation points.

D.2 Profile and code rate

Define the binomial lower-tail polynomial $\tau_{r,k}(x) := \sum_{i=0}^{k-1} \binom{r-1}{i} x^i (1-x)^{r-1-i}$. Inserting (150) into the Bernstein law gives $\gamma(x) = m\, \tau_{r,k}(x)$, whose integral is

$$\int_0^1 \tau_{r,k}(x)\, dx = \sum_{i=0}^{k-1} \binom{r-1}{i} \frac{i!\,(r-1-i)!}{r!} = \sum_{i=0}^{k-1} \frac{1}{r} = \rho, \qquad \rho := \frac{k}{r}, \tag{151}$$

the same result following from dividing the km bits per block by r questions. For a pure code prior $g(x) = m \int_0^x \tau_{r,k}$ and

$$\boxed{L_{\text{code}}(x) = \frac{1-(1-x)\tau_{r,k}(x)}{\int_0^x \tau_{r,k}(t)\, dt}, \qquad L_{\text{code}}(1) = \frac{1}{\rho} = \frac{r}{k}.} \tag{152}$$

For $A = 4$, $r = 4$ and $k = 2$, a random affine polynomial over $\mathbb{F}_4$ gives 16 codewords, any two answers determining the other two, and $\tau_{4,2}(x) = (1-x)^3 + 3x(1-x)^2 = 1 - 3x^2 + 2x^3$, the threshold component of the explicit non-monotonic example (117).

D.3 The secondary large-block limit

Now take $r \to \infty$ with $k/r \to \rho$, *after* taking $n \to \infty$ at fixed r. By the law of large numbers a $\mathrm{Bin}(r-1, x)$ count divided by $r-1$ converges in probability to x, so away from $x = \rho$, $\tau_{r,k}(x) \to \mathbf{1}[x < \rho]$ and hence

$$L_{\text{code}}(x) \longrightarrow \begin{cases} 1, & x < \rho, \\ 1/\rho, & x > \rho. \end{cases} \tag{153}$$

At finite r the transition is smooth and $\gamma(x) > 0$ for every $x < 1$; the jump appears only in the secondary limit, and a Reed–Solomon realization of it requires $m \ge \log_2 r$ to grow.

D.4 Dilution with fresh questions

Let a fraction w of questions lie in code blocks and $1-w$ be independent uniform answers, so that $\gamma(x)/m = w\tau_{r,k}(x) + (1-w)$ and $g(x)/m = w \int_0^x \tau_{r,k} + (1-w)x$; since the initial marginal entropy is m, the leverage is (126). For $1 \le k < r$ differentiating the binomial tail gives $\tau'_{r,k}(x) = -(r-1)\binom{r-2}{k-1} x^{k-1}(1-x)^{r-1-k}$, so

$$\boxed{-(1-x)\gamma'(x) = w\, m\,(r-1) \binom{r-2}{k-1} x^{k-1}(1-x)^{r-k}} \tag{154}$$

is proportional to a $\mathrm{Beta}(k, r-k+1)$ density, with mode $(k-1)/(r-1)$ when $k > 1$ and total mass $wm(1-k/r)$. The release occurs near the code threshold, and fresh questions continue to add entropy at marginal leverage one afterwards, which creates the declining side of the peak. At $k = r$ the code is the full product space, $\tau_{r,r} \equiv 1$, and the deduction density vanishes identically.

D.5 A lapsed MDS block

Let $\mathcal{K} + y \subset \mathbb{F}_A^r$ be an MDS coset with fixed offset y. For $0 \le \delta \le 1$ draw uniformly from the coset with probability $1-\delta$ and uniformly from all A^r answer strings with probability δ, which has full support for every $0 < \delta \le 1$. For $j \le k$ both mixture components project to the uniform distribution on $\mathbb{F}_A^j$, so $H_\delta(j) = jm$. For $j \ge k$ the projection of the code has A^k consistent patterns, each of probability $(1-\delta)A^{-k} + \delta A^{-j} = A^{-k}\big[(1-\delta) + \delta A^{k-j}\big]$, while the other $A^j - A^k$ patterns each have probability δA^{-j}. Summing the two entropy contributions,

$$\boxed{H_\delta(j) = \Lambda\big[km - \log_2 \Lambda\big] + \delta\big(1 - A^{k-j}\big)\big[jm - \log_2 \delta\big], \qquad \Lambda := (1-\delta) + \delta A^{k-j},} \tag{155}$$

for $j \geq k$, with $\eta_\delta(i) = H_\delta(i+1) - H_\delta(i)$. It equals m below threshold; above threshold it declines toward a positive lapse tail, and the exact finite differences should be used rather than replacing the whole tail immediately by δm.

E Local noise as a term-by-term perturbation

E.1 A general channel construction

Let $P_\mathcal{V}$ be any law on $\mathcal{A}^{|\mathcal{V}|}$ and, for $0 \leq \delta \leq 1$, let $\Pr_\delta(y \mid a)$ be a full-support channel on the answer alphabet tending to the identity as $\delta \to 0$. Applying it independently to each coordinate,

$$P_{\mathcal{V},\delta}(y_\mathcal{V}) = \sum_{a_\mathcal{V} \in \mathcal{A}^{|\mathcal{V}|}} P_\mathcal{V}(a_\mathcal{V}) \prod_{q \in \mathcal{V}} \Pr_\delta(y_q \mid a_q), \tag{156}$$

or alternatively using the lapse (127), the full map law remains a product over blocks and has full support if each noisy block law does.

E.2 Why vanishing local noise preserves the limit

For fixed $|\mathcal{V}|$ the block probability simplex is finite-dimensional, so $\|P_{\mathcal{V},\delta} - P_\mathcal{V}\|_{\mathrm{TV}} \to 0$ as $\delta \to 0$. Entropy is continuous on a finite alphabet, so every subset entropy and increment converges, $H_{\mathcal{V},\delta}(j) \to H_\mathcal{V}(j)$ and $\eta_{\mathcal{V},\delta}(i) \to \eta_\mathcal{V}(i)$. The finite hypergeometric weights, and their limiting Bernstein weights, are nonnegative and sum to one, so writing $\gamma_{\mathrm{noisy},n}$ and $\gamma_{\mathrm{hard},n}$ for the profiles computed from the noisy and hard coefficients at size n,

$$\sup_{x \in [0,1]} \big|\gamma_{\mathrm{noisy},n}(x) - \gamma_{\mathrm{hard},n}(x)\big| \;\leq\; \sum_{\mathcal{V} \in \mathcal{P}_n} \frac{|\mathcal{V}|}{Q} \max_i \big|\eta_{\mathcal{V},\delta_{\mathcal{V},n}}(i) - \eta_{\mathcal{V},0}(i)\big| \;\longrightarrow\; 0. \tag{157}$$

The integrals converge uniformly as well, and wherever the limiting received density is bounded away from zero, leverage converges. For fixed block sizes continuity makes the last limit automatic if $\max_\mathcal{V} \delta_{\mathcal{V},n} \to 0$; the aggregate condition is the more general statement, and for varying block sizes a uniform condition is needed, model-specific ones such as $r(n)\delta_n \to 0$ for parity blocks being sharper. At fixed $\delta > 0$ none of these perturbations disappears with n: the exact noisy profile remains the thermodynamic limit.

F Arbitrary mixtures and the cocktail

F.1 Symbolic mixture law

Partition the questions into independent component families, type κ occupying fraction w_κ with initial one-question entropy $h_{0,\kappa}$, macroscopic profile γ_κ and accumulated received density $g_\kappa = \int_0^x \gamma_\kappa$. Entropy adds across independent components, so h_0, γ and g are the corresponding w_κ-weighted sums. Since $h_{0,\kappa} - (1-x)\gamma_\kappa(x) = L_\kappa(x) g_\kappa(x)$ for each component, summing gives (130): the mixture weights in leverage are dynamic received-entropy shares $w_\kappa g_\kappa(x)$, not the fixed question shares w_κ.

F.2 Coin component of the cocktail

For the two-coin component take $(\theta_1, \theta_2) = (0.2, 0.7)$ with weights $(\frac{1}{4}, \frac{3}{4})$ and let the same scalar latent bias govern all bits. With $P_{s,z}$ and F_s as in (100) and (105), the initial entropy per bit for an $m = 4$-bit answer is

$$\frac{F_4}{4} = \frac{3.793149840}{4} = 0.948287460, \tag{158}$$

which is *not* $h_2(0.575) = 0.983708263$, because marginalizing the shared latent makes the four bits dependent. After a vanishing-fraction identification layer the entropy per bit is

$$h_{\text{cond,coin}} = \tfrac{1}{4}h_2(0.2) + \tfrac{3}{4}h_2(0.7) = 0.841450198. \tag{159}$$

F.3 Lapsed-code and fresh components

For $A = 16$, $r = 16$, $k = 4$ and $\delta = 0.1$, compute $H_\delta(j)$ from (155), set $\eta_\delta(i) = H_\delta(i+1) - H_\delta(i)$, and normalize the code profile per bit,

$$\gamma_{\text{lapse}}(x) = \frac{1}{4}\sum_{i=0}^{15}\beta_{15,i}(x)\,\eta_\delta(i), \qquad \int_0^1 \gamma_{\text{lapse}} = \frac{H_\delta(16)}{16\cdot 4} = 0.332328056. \tag{160}$$

The fresh component consists of independent bits with bias 0.3, so its per-bit profile and initial entropy are both $h_2(0.3)$.

F.4 Assembling the cocktail

With question fractions $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ the initial per-bit entropy is (131), the profile for $x > 0$ is (132), and

$$g_{\text{cocktail}}(x) = \tfrac{1}{4}h_{\text{cond,coin}}x + \tfrac{1}{2}\int_0^x \gamma_{\text{lapse}} + \tfrac{1}{4}h_2(0.3)\,x, \tag{161}$$

the scale factor m cancelling from $L = [h_{\text{cocktail}} - (1-x)\gamma_{\text{cocktail}}]/g_{\text{cocktail}}$. At the origin the profile's right limit is $\gamma_{\text{cocktail}}(0^+) = \frac{1}{4}(0.841450198) + \frac{1}{2} + \frac{1}{4}h_2(0.3) = 0.930685274$, so (165) gives the exact singular coefficient

$$\lim_{x\downarrow 0} xL(x) = \frac{h_{\text{cocktail}} - \gamma_{\text{cocktail}}(0^+)}{\gamma_{\text{cocktail}}(0^+)} \simeq 0.02870, \tag{162}$$

while at the endpoint $g_{\text{cocktail}}(1) = \frac{1}{4}(0.841450198) + \frac{1}{2}(0.332328056) + \frac{1}{4}h_2(0.3) = 0.596849302$ and $L(1) = 0.957394590/0.596849302 \simeq 1.60408$. Numerical evaluation of the same closed form gives the dip and peak quoted in Section 7.5.

G What can be inferred from an observed curve?

If h_0 and an ideal macroscopic leverage curve $L(x)$ are known, then $g'(x) = \gamma(x)$ and (91) imply the linear differential equation

$$(1-x)g'(x) + L(x)g(x) = h_0, \qquad g(0) = 0, \tag{163}$$

and for an admissible L, solving it for the regular, finite, nonnegative solution recovers the entropy profile $\gamma = g'$. The qualification matters when L has a $1/x$ singularity, since $g(0) = 0$ is then a singular initial condition rather than a standard nonsingular initial-value problem.

The inverse map does not recover a unique prior. Many microscopic distributions share a subset-entropy profile, and finite data introduce further ambiguity; for a trained model, question selection, optimization dynamics, misspecification and calibration also affect the measured curve. The defensible inference is therefore about broad timing classes: a $1/x$ term indicates information learned on a subextensive clock; a flat curve with $L > 1$ is consistent with first-touch duplication, while $L \equiv 1$ is also the independent baseline; a rising curve is consistent with late constraints; and an interior peak is consistent with threshold release followed by continued reception. This is useful structural evidence, not an identification theorem for the real-world prior.

H Endpoints, jumps, and boundary layers

At the end of the run, $L(1) = h_0/g(1) = \lim_{n\to\infty} QG_{n,1}/G_{n,Q}$. At the origin there are two different cases. If $h_0 = \gamma(0^+) > 0$ and γ is right-differentiable, first-order expansion gives

$$L(0^+) = 1 - \frac{\gamma'(0^+)}{\gamma(0^+)}. \tag{164}$$

If instead $h_0 > \gamma(0^+) > 0$, an initial boundary layer has disappeared into $x = 0$ and

$$\boxed{L(x) \;\sim\; \frac{h_0 - \gamma(0^+)}{\gamma(0^+)}\,\frac{1}{x} \qquad (x \downarrow 0),} \tag{165}$$

which is the source of the spike and coin-mixture hyperbolas.

A separate caveat concerns interior thresholds. If γ merely reaches zero continuously, no instantaneous avalanche follows; a macroscopic jump occurs when $\gamma(x_*^-) > \gamma(x_*^+)$, and especially when the profile jumps to zero. The remaining-entropy density then drops by $(1-x_*)\big[\gamma(x_*^-)-\gamma(x_*^+)\big]$ and cumulative leverage can jump as well; the local per-step leverage inside the shrinking transition layer may diverge while the cumulative $L(x)$ need not. Finally, if $G_{n,Q} = o(Q)$ the normalization of Section 7.2 erases the entire received channel, and one should then introduce a microscopic or mesoscopic time scale rather than conclude automatically that leverage has a meaningful infinite value.